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Integral points for multi-norm tori

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Abstract

We construct a finite subgroup of Brauer–Manin obstruction for detecting the existence of integral points on integral models of principle homogeneous spaces of multi-norm tori. Several explicit examples are provided.

Introduction

The integral points on homogeneous spaces of semi-simple and simply connected linear algebraic groups of noncompact type were studied by Borovoi and Rudnick in [2] and by Colliot-Thélène and the second named author in [3] by using the strong approximation theorem and the Brauer–Manin obstruction. Recently, Harari [7] showed that the Brauer–Manin obstruction accounts for the nonexistence of integral points. Colliot-Thélène noticed that a finite subgroup of the Brauer group is enough to account for the nonexistence of integral points by the compactness argument. However, this result is nonconstructive: one does not know which finite subgroup to use and cannot use it to determine the existence of integral points on the specific equations. In this paper, we give some explicit construction for such finite subgroups for multi-norm tori. The paper is also inspired by Colliot-Thélène's suggestion of studying Gauss' idea for determining integers represented by positive definite binary quadratic forms, which is beautifully explained by Cox in [4], from the point of view of Brauer–Manin obstruction. The advantage of using Brauer–Manin obstruction is to provide more perspective. For example, one can determine the solvability of the negative Pell equations by using the class field theory instead of the continued fractional method (the quadratic Diophantine approximation).

The paper is organized as follows. In Section 1, we construct the idele groups which are the so-called $\mathbf{X}$-admissible groups for determining the integral points for some integral model $\mathbf{X}$. In Section 2, we interpret the $\mathbf{X}$-admissible subgroup in terms of finite Brauer–Manin obstruction and also explain that there is no finite Brauer–Manin obstruction to detect all separated integral models of finite type. In Section 3, we apply our construction to study the classical binary quadratic Diophantine equations. Such classical quadratic Diophantine equations have been studied for a long time by various methods. This construction is the natural extension of Gauss' method for determining integers represented by a positive definite integral binary quadratic form and one can further determine all integers represented by a given binary inhomogeneous quadratic Diophantine equation. In Section 4, we provide several examples of 1-dimensional non-split tori where the splitting fields are imaginary quadratic fields. In Section 5, some examples of 1-dimensional non-split tori where the splitting fields are real quadratic fields are studied. In Section 6, we explain how to apply our construction to study the high-dimensional multi-norm tori by providing some more explicit examples.

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Notation and terminology are standard if not explained. Let F be a number field, $\mathfrak{o}_F$ be the ring of integers of F, Ω_F be the set of all primes in F and ∞_F be the set all infinite primes in F. For simplicity, we write $\mathfrak{p} < \infty_F$, for $\mathfrak{p} \in \Omega_F \setminus \infty_F$. Let $F_\mathfrak{p}$ be the completion of F at $\mathfrak{p}$ and $\mathfrak{o}_{F_\mathfrak{p}}$ be the local completion of $\mathfrak{o}_F$ at $\mathfrak{p}$, for each $\mathfrak{p} \in \Omega_F$. Write $\mathfrak{o}_{F_\mathfrak{p}} = F_\mathfrak{p}$, for $\mathfrak{p} \in \infty_F$. For any positive integer, we use $(F^\times)^k$ (resp. $(F_\mathfrak{p}{}^\times)^k$) to denote the subgroup of kth powers in $F^\times$ (resp. $F_\mathfrak{p}{}^\times$). We also denote the adele (resp. the idele) of F by $\mathbb{A}_F$ (resp. $\mathbb{I}_F$) and

$$F_\infty = \prod_{\mathfrak{p}\in\infty_F} F_\mathfrak{p}.$$

Let $E_1, \ldots, E_n$ be the finite extensions of F and

$$\varphi : R_{E_1/F}(\mathbb{G}_m) \times \ldots \times R_{E_n/F}(\mathbb{G}_m) \longrightarrow \mathbb{G}_m$$

be the homomorphism of algebraic groups which represents

$$(E_1 \otimes_F A)^\times \times \ldots \times (E_n \otimes_F A)^\times \longrightarrow A^\times; (x_1, \ldots, x_n) \mapsto \prod_{i=1}^n N_{E_i/F}(x_i),$$

for any F-algebra A. Define $G := \ker\varphi$ which is called a multi-norm torus over F. Let $\hat{G} = \mathrm{Hom}_{\bar{F}}(G, \mathbb{G}_m)$ be the characters of G where $\bar{F}$ is an algebraic closure of F and $\mathbf{X}$ be a separated $\mathfrak{o}_F$-scheme of finite type whose generic fiber X_F is a principal homogeneous space of G. The obvious necessary conditions for $\mathbf{X}(\mathfrak{o}_F) \neq \emptyset$ are

$$\prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) \neq \emptyset \quad \text{and} \quad X_F(F) \neq \emptyset \tag{A}$$

which is assumed throughout this paper. It should be pointed out that the determination of $X_F(F) \neq \emptyset$ or not is well understood (see [1]).

The Brauer group $\mathrm{Br}(X_F)$ of X_F is defined as

$$\mathrm{Br}(X_F) = H^2_{et}(X_F, \mathbb{G}_m) \quad \text{and} \quad \mathrm{Br}_1(X_F) = \ker[\mathrm{Br}(X_F) \longrightarrow \mathrm{Br}(\bar{X})],$$

where $\bar{X} = X_F \times_F \bar{F}$. Since the image of $\mathrm{Br}(F)$ induced by the structure morphism lies in $\mathrm{Br}_1(X_F)$, one defines

$$\mathrm{Br}_a(X_F) = \mathrm{coker}[\mathrm{Br}(F) \longrightarrow \mathrm{Br}_1(X)].$$

For any subgroup $\mathfrak{s}$ of $\mathrm{Br}_a(X_F)$, one can define the integral Brauer–Manin set with respect to $\mathfrak{s}$ as (see [3])

$$\left(\prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})\right)^\mathfrak{s} = \left\{(x_\mathfrak{p}) \in \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) : \sum_{\mathfrak{p}\in\Omega_F} \mathrm{inv}_\mathfrak{p}(s(x_\mathfrak{p})) = 0,\ \forall s \in \mathfrak{s}\right\}.$$

1. *Construction of* $\mathbf{X}$*-admissible groups*

Since $\mathbf{X}$ is separated over $\mathfrak{o}_F$, one can view $\mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})$ as an open subset of $X_F(F_\mathfrak{p})$ by the natural map, for any $\mathfrak{p} < \infty_F$, and $G(F_\mathfrak{p})$ acts on $X_F(F_\mathfrak{p})$ continuously.

Definition 1.1. Let

$$\mathrm{Stab}(\mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})) = \{g \in G(F_\mathfrak{p}) : g\mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) = \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})\},$$

for $\mathfrak{p} < \infty_F$ and

$$\mathrm{Stab}(\mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})) = G(F_\mathfrak{p}) \quad \text{for } \mathfrak{p} \in \infty_F.$$

Define

$$\mathrm{Stab}_{\mathbb{A}}(\mathbf{X}) = G(\mathbb{A}_F) \cap \left[\prod_{\mathfrak{p} \in \Omega_F} \mathrm{Stab}(\mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})) \right].$$

One has the following basic property of $\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})$.

LEMMA 1.2. *$\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})$ is an open subgroup of $G(\mathbb{A}_F)$.*

Proof. Since the stabilizer of an open subset is open by the continuality, one obtains that $\mathrm{Stab}(\mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}))$ is an open subgroup of $G(F_\mathfrak{p})$, for all $\mathfrak{p} \in \Omega_F$.

Let S be a finite subset of Ω_F containing ∞_F such that G has a model $\mathbf{G}$ over $\mathfrak{o}_S$ and the morphism $G \times_F X_F \to X_F$ can be extended to $\mathbf{G} \times_{\mathfrak{o}_S} \mathbf{X}_S \to \mathbf{X}_S$ where $\mathfrak{o}_S$ is the S-integers of F and $\mathbf{X}_S = \mathbf{X} \times_{\mathfrak{o}_F} \mathfrak{o}_S$. This implies that

$$\mathrm{Stab}(\mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})) \supseteq \mathbf{G}(\mathfrak{o}_{F_\mathfrak{p}}) \quad \text{for all } \mathfrak{p} \notin S.$$

Therefore one concludes that $\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})$ is an open subgroup of $G(\mathbb{A}_F)$. □

Since G is the subgroup of $R_{E_1/F}(\mathbb{G}_m) \times \ldots \times R_{E_n/F}(\mathbb{G}_m)$ by the definition of G, one has the injective map

$$\lambda_{(E_1,\ldots,E_n)} : G(A) \longrightarrow (E_1 \otimes_F A)^\times \times \ldots \times (E_n \otimes_F A)^\times,$$

for any F-algebra A.

DEFINITION 1.3. An open subgroup Ξ of $\prod_{i=1}^n \mathbb{I}_{E_i}$ is called $\mathbf{X}$-admissible, if

$$\lambda_{(E_1,\ldots,E_n)}[\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})] \subseteq \Xi$$

and the map induced by $\lambda_{(E_1,\ldots,E_n)}$

$$\lambda_{(E_1,\ldots,E_n)} : G(\mathbb{A}_F)/G(F)\mathrm{Stab}_{\mathbb{A}}(\mathbf{X}) \longrightarrow \prod_{i=1}^n \mathbb{I}_{E_i} / \left(\prod_{i=1}^n E_i^\times \right) \cdot \Xi$$

is injective.

Since X_F is a trivial G-torsor over F, one can fix an isomorphism $X_F \cong G$ induced by a rational point P. Combining with $\lambda_{(E_1,\ldots,E_n)}$, one can define the injective map

$$f_{(E_1,\ldots,E_n)} : X_F(A) \cong G(A) \xrightarrow{\lambda_{(E_1,\ldots,E_n)}} \prod_{i=1}^n (E_i \otimes_F A)^\times,$$

for any F-algebra A.

Since $\mathbf{X}$ is separated over $\mathfrak{o}_F$, one can regard $\prod_{\mathfrak{p} \in \Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})$ as the subset of $X_F(\mathbb{A}_F)$ by the natural morphism. Then

$$f_{(E_1,\ldots,E_n)} \left[\prod_{\mathfrak{p} \in \Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) \right] \subseteq \prod_{i=1}^n \mathbb{I}_{E_i}.$$

The following proposition explains the reason for introducing the concept of $\mathbf{X}$-admissible subgroups.

PROPOSITION 1.4. *Let Ξ be an $\mathbf{X}$-admissible subgroup of $\prod_{i=1}^n \mathbb{I}_{E_i}$. Then*

$$\mathbf{X}(\mathfrak{o}_F) \neq \emptyset \text{ if and only if } f_{(E_1,\dots,E_n)}\left[\prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})\right] \cap \left[\left(\prod_{i=1}^n E_i^\times\right)\cdot \Xi\right] \neq \emptyset.$$

Proof. Since $\mathbf{X}$ is separated over $\mathfrak{o}_F$, one has $\mathbf{X}(\mathfrak{o}_F) \subseteq \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})$, for all $\mathfrak{p} \in \Omega_F$ and

$$\mathbf{X}(\mathfrak{o}_F) \subseteq \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})$$

by the diagonal map. If $\mathbf{X}(\mathfrak{o}_F) \neq \emptyset$, then

$$f_{(E_1,\dots,E_n)}\left[\prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})\right] \cap \left[\left(\prod_{i=1}^n E_i^\times\right)\cdot \Xi\right] \supseteq f_{(E_1,\dots,E_n)}[\mathbf{X}(\mathfrak{o}_F)] \cap \left(\prod_{i=1}^n E_i^\times\right) \neq \emptyset$$

and the necessity follows.

Conversely, assume that

$$y_A \in \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) \text{ such that } f_{(E_1,\dots,E_n)}(y_A) \in \left(\prod_{i=1}^n E_i^\times\right)\cdot \Xi.$$

By Definition 1.3, there are $\varrho \in G(F)$ and $\sigma_A \in \mathrm{Stab}_{\mathbb{A}}(\mathbf{X})$ such that $y_A = \varrho\sigma_A(P)$ by injectivity of $f_{(E_1,\dots,E_n)}$. This implies that

$$\varrho(P) = \sigma_A^{-1}(y_A) \in \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}).$$

Therefore, $\varrho(P) \in \mathbf{X}(\mathfrak{o}_F) \neq \emptyset$ and the proof is complete. □

If Ξ is an $\mathbf{X}$-admissible subgroup of $\prod_{i=1}^n \mathbb{I}_{E_i}$, there is an open subgroup Ξ_i of $\mathbb{I}_{E_i}$, for each $1 \leqslant i \leqslant n$, such that $\prod_{i=1}^n \Xi_i \subseteq \Xi$. By the class field theory, there is a finite abelian extension K_{Ξ_i}/E_i such that the Artin map

$$\psi_{K_{\Xi_i}/E_i} : \mathbb{I}_{E_i}/E_i^\times \Xi_i \cong \mathrm{Gal}(K_{\Xi_i}/E_i) \tag{1.5}$$

gives the isomorphism for $1 \leqslant i \leqslant n$. Projecting the image of $f_{(E_1,\dots,E_n)}$ to $\mathbb{I}_{E_i}$, one can define

$$f_{E_i} : \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) \longrightarrow \prod_{j=1}^n \mathbb{I}_{E_j} \longrightarrow \mathbb{I}_{E_i},$$

for $1 \leqslant i \leqslant n$.

COROLLARY 1.6. *With the notation as above, $\mathbf{X}(\mathfrak{o}_F) \neq \emptyset$ if and only if there is*

$$x_A \in \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) \text{ such that } \psi_{K_{\Xi_i}/E_i}(f_{E_i}(x_A)) = 1$$

in $\mathrm{Gal}(K_{\Xi_i}/E_i)$*, for all* $1 \leqslant i \leqslant n$.

Proof. Since $\psi_{K_{\Xi_i}/E_i}(f_{E_i}(x_A)) = 1$, for $1 \leqslant i \leqslant n$, one has

$$f_{(E_1,\dots,E_n)}(x_A) \in \left(\prod_{i=1}^n E_i^\times\right)\cdot\left(\prod_{i=1}^n \Xi_i\right) \subseteq \left(\prod_{i=1}^n E_i^\times\right)\cdot \Xi.$$

Then the result follows from the same argument as those in Proposition 1.4. □

The main purpose of this section is to construct such $\mathbf{X}$-admissible groups. First, we need to the following lemma.

LEMMA 1.7. *If $C_{\mathfrak{p}}$ is an open subgroup of $G(F_{\mathfrak{p}})$ and k is a positive integer, then*

$$\lambda_{(E_1,\dots,E_n)}(C_{\mathfrak{p}}) \cdot \{(a,\dots,a) : a \in (F_{\mathfrak{p}}^{\times})^k\}$$

is an open subgroup of $\prod_{i=1}^{n} E_{i\mathfrak{p}}^{\times}$ for $\mathfrak{p}$-adic topology with $\mathfrak{p} < \infty_F$.

Proof. Since $C_{\mathfrak{p}}$ is an open subgroup of $G(F_{\mathfrak{p}})$, there are positive integers $a_1,\dots,a_n$ such that

$$\left[\prod_{i=1}^{n}(1+\mathfrak{p}^{a_i}\mathfrak{o}_{E_{i\mathfrak{p}}})\right] \cap \lambda_{(E_1,\dots,E_n)}(G(F_{\mathfrak{p}})) \subseteq \lambda_{(E_1,\dots,E_n)}(C_{\mathfrak{p}}).$$

By Hensel's lemma, there is a sufficiently large positive integer $b_i > a_i$ such that

$$(1+\mathfrak{p}^{b_i}\mathfrak{o}_{E_{i\mathfrak{p}}}) \subseteq (1+\mathfrak{p}^{a_i}\mathfrak{o}_{E_{i\mathfrak{p}}})^{lk},$$

where $l = \sum_{i=1}^{n} l_i$ and $l_i = [E_i : F]$, for all $1 \leqslant i \leqslant n$.

If

$$(x_i)_{i=1}^{n} \in \prod_{i=1}^{n}(1+\mathfrak{p}^{b_i}\mathfrak{o}_{E_{i\mathfrak{p}}}),$$

then

$$(x_i)_{i=1}^{n} = (y_i^{lk})_{i=1}^{n} \quad \text{with } (y_i)_{i=1}^{n} \in \prod_{i=1}^{n}(1+\mathfrak{p}^{a_i}\mathfrak{o}_{E_{i\mathfrak{p}}}).$$

Since

$$y_i^{lk} = \left[y_i^{lk}\left(\prod_{i=1}^{n} N_{E_i/F}(y_i)\right)^{-k}\right] \cdot \left[\prod_{i=1}^{n} N_{E_i/F}(y_i)\right]^{k}$$

and

$$\left(\left[y_i^{lk}\left(\prod_{i=1}^{n} N_{E_i/F}(y_i)\right)^{-k}\right]\right)_{i=1}^{n} \in \lambda_{(E_1,\dots,E_n)}(C_{\mathfrak{p}}),$$

one concludes that

$$\prod_{i=1}^{n}(1+\mathfrak{p}^{b_i}\mathfrak{o}_{E_{i\mathfrak{p}}}) \subseteq \lambda_{(E_1,\dots,E_n)}(C_{\mathfrak{p}}) \cdot \{(a,\dots,a) : a \in (F_{\mathfrak{p}}^{\times})^k\}$$

and the proof is complete. □

In order to prove the existence of $\mathbf{X}$-admissible groups, we further need the following lemma.

LEMMA 1.8. *Let k be a positive integer. Then there is a finite subset T_0 of $\Omega_F \setminus \infty_F$ such that*

$$F^{\times} \cap \left[\prod_{\mathfrak{p}\in T}(F_{\mathfrak{p}}^{\times})^{2kh_F} \times \prod_{\mathfrak{p}\notin T}\mathfrak{o}_{F_{\mathfrak{p}}}^{\times}\right] \subseteq (F^{\times})^k,$$

for any $T \supseteq T_0$, where h_F is the class number of F.

Proof. Since $\mathfrak{o}_F^\times/(\mathfrak{o}_F^\times)^{2k}$ is finite by the Dirichlet unit theorem, there are only finitely many cosets $\alpha_i(\mathfrak{o}_F^\times)^{2k}$ of $\mathfrak{o}_F^\times/(\mathfrak{o}_F^\times)^{2k}$ such that $\alpha_i \notin (F_{\mathfrak{p}_i}^\times)^{2k}$, for some prime $\mathfrak{p}_i < \infty_F$. Fix one such prime $\mathfrak{p}_i$ for each $\alpha_i(\mathfrak{o}_F^\times)^{2k}$ and define T_0 to be the set consisting of all such $\mathfrak{p}_i$. It is possible that T_0 is empty.

Let

$$\alpha \in F^\times \cap \left[\prod_{\mathfrak{p}\in T}(F_{\mathfrak{p}}^\times)^{2kh_F} \times \prod_{\mathfrak{p}\notin T}\mathfrak{o}_{F_{\mathfrak{p}}}^\times\right],$$

for any finite set $T \supseteq T_0$. Since $\mathfrak{p}^{h_F}$ is a principal ideal, there is $\varpi_{\mathfrak{p}} \in F$ such that $\mathfrak{p}^{h_F} = \varpi_{\mathfrak{p}}\mathfrak{o}_F$, for each finite $\mathfrak{p} \in T$. There is $t_{\mathfrak{p}} \in \mathbb{Z}$, for each finite $\mathfrak{p} \in T$ such that

$$\beta = \alpha\left(\prod_{\mathfrak{p}\in T\setminus\infty_F}\varpi_{\mathfrak{p}}^{t_{\mathfrak{p}}}\right)^{2k} \in \mathfrak{o}_F^\times.$$

Then $\beta(\mathfrak{o}_F^\times)^{2k}$ is not one of the above-mentioned cosets $\alpha_i(\mathfrak{o}_F^\times)^{2k}$. This implies that

$$\alpha \in F^\times \cap \left[\prod_{\mathfrak{p}\in\infty_F}F_{\mathfrak{p}}^\times \times \prod_{\mathfrak{p}\notin\infty_F}(F_{\mathfrak{p}}^\times)^{2k}\right].$$

Therefore,

$$F^\times \cap \left[\prod_{\mathfrak{p}\in T}(F_{\mathfrak{p}}^\times)^{2kh_F} \times \prod_{\mathfrak{p}\notin T}\mathfrak{o}_{F_{\mathfrak{p}}}^\times\right] \subseteq F^\times \cap \left[\prod_{\mathfrak{p}\in\infty_F}F_{\mathfrak{p}}^\times \times \prod_{\mathfrak{p}\notin\infty_F}(F_{\mathfrak{p}}^\times)^{2k}\right].$$

Let

$$x \in F^\times \cap \left[\prod_{\mathfrak{p}\in\infty_F}F_{\mathfrak{p}}^\times \times \prod_{\mathfrak{p}\notin\infty_F}(F_{\mathfrak{p}}^\times)^{2k}\right].$$

Applying (9.1.3) Theorem in [11] for μ_{2k}, one obtains that $x^2 \in (F^\times)^{2k}$. There is $y \in F^\times$ such that $x = y^k$ or $x = -y^k$.

Suppose $x = -y^k$. Let $k = 2^s k_1$ with $2 \nmid k_1$ and $\zeta_{2^{s+1}}$ be a primitive 2^{s+1}th roots of unity. Then $\zeta_{2^{s+1}} \in F_{\mathfrak{p}}^\times$, for all $\mathfrak{p} < \infty_F$. By the Chebotarev density theorem, one concludes that $\zeta_{2^{s+1}} \in F$. Therefore, $-1 = (\zeta_{2^{s+1}})^k \in (F^\times)^k$ and the proof is complete. □

The proof of Lemma 1.8 also provides the explicit method to determine the set T_0. For example, when $F = \mathbb{Q}$, one has $T_0 = \{2\}$ or $\{p\}$ with $p \equiv 3 \bmod 4$. The following corollary has its own independent interest.

COROLLARY 1.9. *For any open subgroup Ξ of $\mathbb{I}_F$ and a positive integer k, there is an open subgroup Ξ_k of $\mathbb{I}_F$ such that $(F^\times \cap \Xi_k) \subseteq (F^\times \cap \Xi)^k$.*

Proof. Let $\Xi_{\mathfrak{p}}$ be an open subgroup of $F_{\mathfrak{p}}$ for each $\mathfrak{p} \in \Omega_F$ such that $\prod_{\mathfrak{p}\in\Omega_F}\Xi_{\mathfrak{p}} \subseteq \Xi$ and $\Xi_{\mathfrak{p}} = \mathfrak{o}_{F_{\mathfrak{p}}}^\times$ for almost all $\mathfrak{p} \in \Omega_F$. Let S_0 be a finite subset of Ω_F such that $\Xi_{\mathfrak{p}} = \mathfrak{o}_{F_{\mathfrak{p}}}^\times$ for all $\mathfrak{p} \notin S_0$. For each $\mathfrak{p} \in S_0$ and for a prime divisor q of k, one defines $t_{\mathfrak{p}}(q)$ to be the maximal positive integer t such that the primitive q^t-th roots of unity are in $F_{\mathfrak{p}}$. Let

$$t_{\mathfrak{p}} := \max\{t_{\mathfrak{p}}(q) : \text{ prime divisor } q \text{ of } k\} \quad \text{and} \quad r := \prod_{\mathfrak{p}\in S_0}(t_{\mathfrak{p}}+1).$$

Applying Lemma 1.8 for k^{r+1}, one obtains a finite subset T_0 of Ω_F. Let $S = S_0 \cup T_0$ and defines

$$\Xi_k = \prod_{\mathfrak{p}\in S} \Xi_{\mathfrak{p}}^{2h_F k^{r+1}} \times \prod_{\mathfrak{p}\notin S} \Xi_{\mathfrak{p}}.$$

For any $x \in (F^\times \cap \Xi_k)$, there is $y \in F^\times$ such that $x = y^{k^{r+1}}$ by Lemma 1.8. Moreover, there is $\xi_{\mathfrak{p}} \in \Xi_{\mathfrak{p}}$ such that

$$x = y^{k^{r+1}} = \xi_{\mathfrak{p}}^{k^{r+1}},$$

for all $\mathfrak{p} \in S$.

If $\mathfrak{p} \in S_0$, one has that

$$y^{k^r} = \xi_{\mathfrak{p}}^{k^r} \in \Xi_{\mathfrak{p}}$$

by the maximality of $t_{\mathfrak{p}}$.

If $\mathfrak{p} \notin S_0$, one has

$$y^{k^r} \in \mathfrak{o}_{F_{\mathfrak{p}}}^\times = \Xi_{\mathfrak{p}}.$$

Therefore, $y^{k^r} \in (F^\times \cap \Xi)$ and the proof is complete. □

Now we can show the existence of the admissible subgroups for $\mathbf{X}$.

THEOREM 1.10. *The* $\mathbf{X}$*-admissible subgroups of* $\prod_{i=1}^n \mathbb{I}_{E_i}$ *always exist.*

Proof. Since X_F is a trivial torsor over F under G, there is a finite subset $S_0 \subset (\Omega_F \setminus \infty_F)$ such that:

(1) the map φ can be extended to

$$\Phi : R_{\mathfrak{o}_{E_1,S_0}/\mathfrak{o}_{F,S_0}}(\mathbb{G}_m) \times \ldots \times R_{\mathfrak{o}_{E_n,S_0}/\mathfrak{o}_{F,S_0}}(\mathbb{G}_m) \longrightarrow \mathbb{G}_m$$

over $\mathfrak{o}_{F,S_0}$, where $\mathfrak{o}_{F,S_0}$, and $\mathfrak{o}_{E_1,S_0}, \ldots \mathfrak{o}_{E_n,S_0}$ are the S_0-integers of F and $E_1, \ldots, E_n$, respectively. Moreover, $\ker(\Phi) = \mathbf{G}$ is a model of G over $\mathfrak{o}_{F,S_0}$.

(2) $\mathbf{X}_{S_0}$ is a trivial torsor over $\mathfrak{o}_{S_0}$ under $\mathbf{G}$, where $\mathbf{X}_{S_0} = \mathbf{X} \times_{\mathfrak{o}_F} \mathfrak{o}_{F,S_0}$.

This implies that

$$\lambda_{(E_1,\ldots,E_n)}[\mathrm{Stab}(\mathbf{X}(\mathfrak{o}_{F_{\mathfrak{p}}}))] = \ker\left(\prod_{i=1}^n N_{E_i/F} : \prod_{i=1}^n \mathfrak{o}_{E_{i\mathfrak{p}}}^\times \longrightarrow \mathfrak{o}_{F_{\mathfrak{p}}}^\times\right),$$

for every $\mathfrak{p} \notin S_0$.

Let $l = \sum_{i=1}^n l_i$ with $l_i = [E_i : F]$, for all $1 \leqslant i \leqslant n$. For each $\mathfrak{p} \in S_0$ and prime divisor q of l, one defines $t_{\mathfrak{p}}(q)$ to be the maximal positive integer t such that the primitive q^tth roots of unity are in $F_{\mathfrak{p}}$. Let

$$t_{\mathfrak{p}} := \max\{t_{\mathfrak{p}}(q) : \ q \text{ divides } l\} \quad \text{and} \quad r := \prod_{\mathfrak{p}\in S_0} (t_{\mathfrak{p}} + 1).$$

Let T_0 be a finite subset of Ω_F outside ∞_F such that Lemma 1.8 holds for $k = l^{r+1}$. Put $S = S_0 \cup T_0$ and

$$\Xi = \left[\prod_{\mathfrak{p}\in S} \lambda_{(E_1,\ldots,E_n)}(\mathrm{Stab}(\mathbf{X}(\mathfrak{o}_{F_{\mathfrak{p}}}))) \cdot H_{\mathfrak{p}}\right] \times \left(\prod_{i=1}^n \prod_{\mathfrak{p}\notin S} \mathfrak{o}_{E_{i\mathfrak{p}}}^\times\right),$$

where

$$H_{\mathfrak{p}} = \{(a, \ldots, a) : a \in (F_{\mathfrak{p}}^\times)^{2l^r h_F}\}$$

and h_F is the class number of F. By Lemma 1.7, one has that Ξ is an open subgroup of $\prod_{i=1}^n \mathbb{I}_{E_i}$ and

$$\lambda_{(E_1,\dots,E_n)}[\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})] \subseteq \Xi.$$

Suppose $\sigma \in G(\mathbb{A}_F)$ with $\lambda_{(E_1,\dots,E_n)}(\sigma) = ai$ where

$$a = (a_i)_{i=1}^n \in \prod_{i=1}^n E_i^\times \quad \text{and} \quad i \in \Xi.$$

Then

$$\prod_{i=1}^n N_{E_i/F}(a_i) \in \left[\prod_{\mathfrak{p}\in S} (F_{\mathfrak{p}}^\times)^{2l^{r+1}h_F} \times \prod_{\mathfrak{p}\notin S} \mathfrak{o}_{F_{\mathfrak{p}}}^\times\right].$$

By Lemma 1.8, there is $u \in F^\times$ such that $\prod_{i=1}^n N_{E_i/F}(a_i) = u^{l^{r+1}}$. This implies that

$$(a_1 u^{-l^r}, \dots, a_n u^{-l^r}) \in \lambda_{(E_1,\dots,E_n)}(G(F)).$$

Write $i = (i_{\mathfrak{p}})_{\mathfrak{p}\in\Omega_F}$ and $i_{\mathfrak{p}} = s_{\mathfrak{p}} \cdot (n_{\mathfrak{p}}, \dots, n_{\mathfrak{p}})$ with

$$s_{\mathfrak{p}} \in \lambda_{(E_1,\dots,E_n)}(\mathrm{Stab}(\mathbf{X}(\mathfrak{o}_{F_{\mathfrak{p}}})) \quad \text{and} \quad n_{\mathfrak{p}} \in (F_{\mathfrak{p}}^\times)^{2l^r h_F}$$

for $\mathfrak{p} \in S$. Therefore, $u^{l^{r+1}} n_{\mathfrak{p}}^l = 1$ for $\mathfrak{p} \in S$.

If $\mathfrak{p} \in S_0$, one has that $u^{l^r} n_{\mathfrak{p}} = 1$ by the maximality of $t_{\mathfrak{p}}$.

If $\mathfrak{p} \in S \setminus S_0$, one has

$$u^{l^r}(n_{\mathfrak{p}}, \dots, n_{\mathfrak{p}}) \in \ker\left(\prod_{i=1}^n N_{E_i/F} : \prod_{i=1}^n \mathfrak{o}_{E_{i\mathfrak{p}}}^\times \longrightarrow \mathfrak{o}_{F_{\mathfrak{p}}}^\times\right).$$

Therefore one concludes that

$$u^{l^r} i \in \lambda_{(E_1,\dots,E_n)}[\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})] \quad \text{and} \quad \sigma \in G(F)\,\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})$$

and the proof is complete. □

2. *Finite Brauer–Manin obstruction*

In this section, we interpret the $\mathbf{X}$-admissible subgroup in terms of Brauer–Manin obstruction. We keep the same notation as before. By the definition of G, one has the surjective map

$$\widehat{R_{E_1/F}(\mathbb{G}_m)} \times \ldots \times \widehat{R_{E_n/F}(\mathbb{G}_m)} \longrightarrow \hat{G}$$

which induces

$$\phi : \prod_{i=1}^n H^2(F, \widehat{R_{E_i/F}(\mathbb{G}_m)}) = \prod_{i=1}^n H^2(E_i, \mathbb{Z}) \longrightarrow H^2(F, \hat{G}) \tag{2.1}$$

by Shapiro's lemma (see (1.6.3) Proposition in [11]).

Let Ξ be an $\mathbf{X}$-admissible subgroup of $\prod_{i=1}^n \mathbb{I}_{E_i}$. There is an open subgroup Ξ_i of $\mathbb{I}_{E_i}$, for each $1 \leqslant i \leqslant n$ such that $\prod_{i=1}^n \Xi_i \subseteq \Xi$. By the class field theory, there is a finite abelian extension K_{Ξ_i}/E_i satisfying (1.5), for $1 \leqslant i \leqslant n$.

By the short exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

with the trivial action of $\mathrm{Gal}(K_{\Xi_i}/E_i)$, one obtains an isomorphism of finite groups

$$\delta_i : \mathrm{Hom}(\mathrm{Gal}(K_{\Xi_i}/E_i), \mathbb{Q}/\mathbb{Z}) \cong H^2(\mathrm{Gal}(K_{\Xi_i}/E_i), \mathbb{Z}). \tag{2.2}$$

Let $b_i(\Xi_i)$ be the image of the inflation

$$H^2(\mathrm{Gal}(K_{\Xi_i}/E_i),\mathbb{Z}) \longrightarrow H^2(E_i,\mathbb{Z}), \tag{2.3}$$

for $1 \leqslant i \leqslant n$. Since $X_F \cong G$ by the fixed rational point P, one has

$$\rho : H^2(F,\hat{G}) \cong \mathrm{Br}_1(G) \cong \mathrm{Br}_1(X_F)$$

by [18, Theorem 1 of § 4.3]. Let $b(\Xi)$ be the subgroup of $\mathrm{Br}_1(X_F)$ generated by the image of $\{b_i(\Xi_i)\}_{i=1}^n$ under the map $\rho\circ\phi$. It is clear that $b(\Xi)$ is finite.

Proposition 2.4. *With the above notation,*

$$\mathbf{X}(\mathfrak{o}_F) \neq \emptyset \ \text{ if and only if } \left[\prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})\right]^{b(\Xi)} \neq \emptyset.$$

Proof. One only needs to show the sufficiency. For any

$$\chi \in \mathrm{Hom}(\mathrm{Gal}(K_{\Xi_i}/E_i),\mathbb{Q}/\mathbb{Z}) \quad \text{and} \quad (x_\mathfrak{p})_{\mathfrak{p}\in\Omega_F} \in \left[\prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})\right]^{b(\Xi)},$$

there is $(\sigma_\mathfrak{p})_{\mathfrak{p}\in\Omega_F} \in G(\mathbb{A}_F)$ such that

$$f_{(E_1,\dots,E_n)}((x_\mathfrak{p})_{\mathfrak{p}\in\Omega_F}) = \lambda_{(E_1,\dots,E_n)}((\sigma_\mathfrak{p})_{\mathfrak{p}\in\Omega_F}).$$

Since

$$H^1(K_{\Xi_i},\mathbb{Z}) \cong \mathrm{Hom}_{\mathrm{cont}}(\mathrm{Gal}(\bar{E}/K_\Xi),\mathbb{Z}) = 0,$$

the inflation map in (2.3) is injective. One can view $\delta_i(\chi) \in b_i(\Xi_i)$ and obtain

$$\begin{aligned} 0 &= \sum_{\mathfrak{p}\in\Omega_F} \mathrm{inv}_\mathfrak{p}(\rho\circ\phi(\delta_i(\chi))(x_\mathfrak{p})) = \sum_{\mathfrak{p}\in\Omega_F} \mathrm{inv}_\mathfrak{p}(\phi(\delta_i(\chi))(\sigma_\mathfrak{p})) \\ &= \sum_{\mathfrak{p}\in\Omega_F} \mathrm{inv}_\mathfrak{p}(\delta_i(\chi)\cup(\lambda_{(E_1,\dots,E_n)}(\sigma_\mathfrak{p}))) = \chi(\psi_{K_{\Xi_i}/E_i}(\lambda_{(E_1,\dots,E_n)}((\sigma_\mathfrak{p})_{\mathfrak{p}\in\Omega_F}))) \end{aligned}$$

by the functoriality of the Brauer–Manin pairing (see [17, (5.3)]), the compatibility of Bauer–Manin pairing with the cup product (see [8, Section 6]) and (8.1.11) Proposition in [11]. This implies

$$\psi_{K_{\Xi_i}/E_i}(f_{E_i}((x_\mathfrak{p})_{\mathfrak{p}\in\Omega_F}) = 0,$$

for all $1 \leqslant i \leqslant n$. The result follows from Corollary 1.6. □

It should be pointed out that the finite subgroup $b(\Xi)$ of $\mathrm{Br}_1(X_F)$ depends on the integral model $\mathbf{X}$. In general, there does not exist a universal finite subgroup $\mathfrak{s}$ of $\mathrm{Br}_1(X_F)$, which can be used to test the existence of the integral points for all integral model $\mathbf{X}$ of X_F.

Example 2.5. Let $F=\mathbb{Q}$ and $E=\mathbb{Q}(\sqrt{-1})$ and

$$G = X_\mathbb{Q} = R^1_{E/F}(\mathbb{G}_m) = \mathrm{Spec}(\mathbb{Q}[x,y]/(x^2+y^2-1)).$$

For any finite subgroup $\mathfrak{s}$ of $\mathrm{Br}_1(X_\mathbb{Q})$, there exists a separated $\mathbb{Z}$-scheme $\mathbf{X}$ of finite type with the generic fiber $X_\mathbb{Q}$ such that

$$\left(\prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p)\right)^{\mathfrak{s}} \neq \emptyset \quad \text{but } \mathbf{X}(\mathbb{Z}) = \emptyset.$$

Proof. Since $(x+\sqrt{-1}y)(x-\sqrt{-1}y)=x^2+y^2$, one has

$$\bar{\mathbb{Q}}[X_{\mathbb{Q}}]^{\times}=\bar{\mathbb{Q}}^{\times}\times f_E^{\mathbb{Z}} \quad \text{with } f_E=x+y\sqrt{-1}.$$

Define

$$\mathrm{Ind}_{\mathrm{Gal}(\bar{\mathbb{Q}}/E)}^{\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})}(f_E^{\mathbb{Z}})\longrightarrow f_E^{\mathbb{Z}},\quad \xi\mapsto\sum_{\sigma\in\mathrm{Gal}(E/\mathbb{Q})}\sigma\xi(\sigma^{-1}).$$

One has the exact sequence of $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-module

$$1\longrightarrow\mathbb{Z}\longrightarrow\mathrm{Ind}_{\mathrm{Gal}(\bar{\mathbb{Q}}/E)}^{\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})}(f_E^{\mathbb{Z}})\longrightarrow f_E^{\mathbb{Z}}\longrightarrow 1.$$

The Galois cohomology gives

$$H^2(\mathbb{Q},\mathrm{Ind}_{\mathrm{Gal}(\bar{\mathbb{Q}}/E)}^{\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})}(f_E^{\mathbb{Z}}))\longrightarrow H^2(\mathbb{Q},f_E^{\mathbb{Z}})\longrightarrow H^3(\mathbb{Q},\mathbb{Z}).$$

By [**10**, Corollary 4.17], one has $H^3(\mathbb{Q},\mathbb{Z})=0$. By Shapiro's lemma (see (1.6.4) Proposition [**11**]), the corestriction map

$$\mathrm{cor}_{E/\mathbb{Q}}:\ H^2(E,f_E^{\mathbb{Z}})\longrightarrow H^2(\mathbb{Q},f_E^{\mathbb{Z}}) \tag{2.6}$$

is surjective. Since $\mathrm{Gal}(\bar{\mathbb{Q}}/E)$ acts on $f_E^{\mathbb{Z}}$ trivially, one gets

$$\delta:\ \mathrm{Hom}(\mathrm{Gal}(\bar{\mathbb{Q}}/E),\mathbb{Q}/\mathbb{Z})\cong H^2(E,f_E^{\mathbb{Z}}) \tag{2.7}$$

by using the exact sequence with the trivial action

$$0\longrightarrow\mathbb{Z}\longrightarrow\mathbb{Q}\longrightarrow\mathbb{Q}/\mathbb{Z}\longrightarrow 0.$$

By [**18**, Theorem 1 of § 4.3], there is a finite subgroup

$$S\subset\mathrm{Hom}(\mathrm{Gal}(\bar{\mathbb{Q}}/E),\mathbb{Q}/\mathbb{Z}) \text{ such that } \mathrm{cor}_{E/\mathbb{Q}}\circ\delta(S)=\mathfrak{s}.$$

By the class field theory, there are a finite abelian extension K_S/E such that

$$S=\{\chi\in\mathrm{Hom}(\mathrm{Gal}(\bar{\mathbb{Q}}/E),\mathbb{Q}/\mathbb{Z}):\chi|_{\mathrm{Gal}(\bar{\mathbb{Q}}/K_S)}=1\}$$

and an open subgroup U of $\mathbb{I}_E$ with

$$\psi_S:\mathbb{I}_E/E^{\times}U\cong\mathrm{Gal}(K_S/E)$$

via the Artin map. Let l be an odd prime such that the invertible elements of the l-adic completion of $\mathfrak{o}_E$ are contained in U. Define $\mathbf{X}$ by the equations

$$\begin{cases}x^2+y^2=1,\\ x-l+l^2z=0.\end{cases}$$

By Hensel's lemma, there is $y_0\in\mathbb{Z}_l$ such that $(1-l)^2+2y_0+l^2y_0^2=0$. Choose the local solution $(s_p)_p\in\prod_{p\leqslant\infty}\mathbf{X}(\mathbb{Z}_p)$ by

$$s_p=(x_p,y_p,z_p)=\begin{cases}(1,0,l^{-2}(l-1)) & \text{if } p\neq l,\\ (l(1-l),1+l^2y_0,1) & \text{if } p=l.\end{cases}$$

Since $f_E(s_p)\in\prod_{\mathfrak{p}|p}E_{\mathfrak{p}}$ and more precisely

$$f_E((x_p,y_p,z_p))=\begin{cases}(x_p+y_p\sqrt{-1},x_p-y_p\sqrt{-1}) & \text{if } p \text{ splits in } E/\mathbb{Q},\\ x_p+y_p\sqrt{-1} & \text{otherwise}\end{cases}$$

for all primes p including ∞, one has $f_E((s_p)_p)\in U$. For any $\beta\in\mathfrak{s}$, there is $\chi\in S$ such that $\beta=\mathrm{cor}_{E/\mathbb{Q}}\circ\delta(\chi)$. By [**11**, (8.1.4) Proposition and (7.1.4) Corollary], one has

$$\sum_{p\leqslant\infty}\mathrm{inv}_p(\beta(s_p))=\sum_{\mathfrak{p}\in\Omega_E}\mathrm{inv}_{\mathfrak{p}}(\delta(\chi)(s_p)).$$

By [11, (8.1.11) Proposition], one concludes that

$$\sum_{\mathfrak{p}\in\Omega_E} \mathrm{inv}_{\mathfrak{p}}(\delta(\chi)(s_p)) = \chi(\psi_S(f_E((s_p)_p)) = 0.$$

This implies that

$$(s_p)_{p\leqslant\infty} \in \left(\prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p)\right)^{\mathfrak{s}}.$$

However, it is clear that $\mathbf{X}(\mathbb{Z}) = \emptyset$. □

3. *Binary quadratic Diophantine equations*

In this section, we apply the results in the previous sections to $\mathfrak{o}_F$-scheme $\mathbf{X}$ defined by the following irreducible polynomial:

$$ax^2 + bxy + cy^2 + ex + fy + g = 0 \tag{3.1}$$

with $a, b, c, e, f, g \in \mathfrak{o}_F$ and $-d = b^2 - 4ac \in F^\times$.

Let $V = Fv_1 + Fv_2$ be a 2-dimensional quadratic space over F with a basis $\{v_1, v_2\}$ and the associated symmetric bilinear map $B : V \times V \to F$ satisfying

$$B(v_1, v_1) = a, \quad B(v_2, v_2) = c \quad \text{and} \quad B(v_1, v_2) = \tfrac{1}{2}b.$$

Let the group G represent $\mathrm{SO}(V \otimes_F A)$, for any F-algebra A. Then X_F is a G-torsor over F and $G(F) = \mathrm{SO}(V)$. Let $\mathrm{SO}_{\mathbb{A}}(V)$ and $GL_{\mathbb{A}}(V)$ be the adelic groups of $\mathrm{SO}(V)$ and $GL(V)$, respectively.

An $\mathfrak{o}_F$-lattice L in V is defined as a finitely generated $\mathfrak{o}_F$-module satisfying $FL = V$ and $L_{\mathfrak{p}}$ is the local completion of L inside the local completion $V_{\mathfrak{p}}$ of V at $\mathfrak{p}$, for $\mathfrak{p} < \infty_F$ and $L_{\mathfrak{p}} = V_{\mathfrak{p}}$, for $\mathfrak{p} \in \infty_F$.

LEMMA 3.2. *Let L be an $\mathfrak{o}_F$-lattice in V and $u_0 \in V$. For any $(\sigma_{\mathfrak{p}})_{\mathfrak{p}\in\Omega_F} \in GL_{\mathbb{A}}(V)$, there is a unique $\mathfrak{o}_F$-lattice L' in V and $u_0' \in V$ unique up to elements in L' such that*

$$L'_{\mathfrak{p}} + u_0' = \sigma_{\mathfrak{p}}(L_{\mathfrak{p}} + u_0)$$

inside $V_{\mathfrak{p}}$, for all $\mathfrak{p} < \infty_F$.

Proof. By [13, 81:14], there is an $\mathfrak{o}_F$-lattice L' of V such that $L'_{\mathfrak{p}} = \sigma_{\mathfrak{p}} L_{\mathfrak{p}}$ for all $\mathfrak{p} < \infty_F$. By the strong approximation theorem for $\mathbb{G}_a$, there is $u_0' \in V$ such that

$$\sigma_{\mathfrak{p}} u_0 - u_0' \in L'_{\mathfrak{p}},$$

for all $\mathfrak{p} < \infty_F$. Therefore,

$$L'_{\mathfrak{p}} + u_0' = L'_{\mathfrak{p}} + \sigma_{\mathfrak{p}} u_0 = \sigma_{\mathfrak{p}} L_{\mathfrak{p}} + \sigma_{\mathfrak{p}} u_0 = \sigma_{\mathfrak{p}}(L_{\mathfrak{p}} + u_0),$$

for all $\mathfrak{p} < \infty_F$ as required.

Suppose there is an $\mathfrak{o}_F$-lattice M in V and $x_0 \in V$ such that

$$x_0 + M_{\mathfrak{p}} = u_0' + L'_{\mathfrak{p}},$$

for all $\mathfrak{p} < \infty_F$. This implies that $M_{\mathfrak{p}} = L'_{\mathfrak{p}}$ and $x_0 - u_0' \in L'_{\mathfrak{p}}$, for all $\mathfrak{p} < \infty_F$. Therefore, $M = L'$ and $x_0 - u_0' \in L'$ and the uniqueness follows. □

By Lemma 3.2, one can define the action of $GL_{\mathbb{A}}(V)$ on $L+u_0$ and set

$$\mathrm{SO}_{\mathbb{A}}(L+u_0)=\{\sigma\in \mathrm{SO}_{\mathbb{A}}(V):\sigma(L+u_0)=(L+u_0)\}.$$

By (A), $X_F(F)\neq\emptyset$ by the Hasse principle. To fix a rational point P in $X_F(F)$ is equivalent to fix a non-zero vector $x_0\in V$ such that

$$B(x_0,x_0)=B(u_0,u_0)-g.$$

In particular, we can choose

$$L=\mathfrak{o}_F v_1+\mathfrak{o}_F v_2 \quad\text{and}\quad u_0=d^{-1}(2ce-bf)v_1+d^{-1}(2af-be)v_2\in V.$$

If $\sigma\in \mathrm{SO}(V)$ with $\sigma x_0\in(L+u_0)$, there are $\alpha,\beta\in\mathfrak{o}_F$ such that $\sigma x_0=\alpha v_1+\beta v_2+u_0$. One can verify that (α,β) is a solution of the equation (3.1). This gives the map

$$\{\sigma\in \mathrm{SO}(V):\sigma x_0\in(L+u_0)\}\longrightarrow \mathbf{X}(\mathfrak{o}_F).$$

In fact, this map is bijective by the Witt theorem [**13**, 42:16 and 42:17]. One can also extend this bijection to

$$\mathbf{X}(\mathfrak{o}_{F_{\mathfrak{p}}})\cong\{\sigma\in \mathrm{SO}(V_{\mathfrak{p}}):x_0\in\sigma(L_{\mathfrak{p}}+u_0)\},$$

for all $\mathfrak{p}<\infty_F$. Since $u_0\in L_{\mathfrak{p}}$ and $\mathfrak{o}_{F_{\mathfrak{p}}}x_0$ splits $L_{\mathfrak{p}}$ for almost all $\mathfrak{p}$, one has

$$\mathrm{SO}_{\mathbb{A}}(L+u_0)\subseteq \mathrm{Stab}_{\mathbb{A}}(\mathbf{X})$$

with finite index. In practice, we replace $\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})$ by $\mathrm{SO}_{\mathbb{A}}(L+u_0)$ which is much easier to be computed.

3.1. *Split cases* $-d=b^2-4ac\in(F^\times)^2$

Fix the isotropic basis of V

$$\{(b+\varpi)v_1-2av_2,(b-\varpi)v_1-2av_2\},$$

where $\varpi\in F^\times$ satisfying $\varpi^2=b^2-4ac$. For any F-algebra A, there is a group isomorphism $\mathrm{SO}(V\otimes_F A)\cong A^\times$ by sending $(\sigma\mapsto\varrho)$ such that

$$\sigma((b+\varpi)v_1-2av_2)=\varrho[(b+\varpi)v_1-2av_2].$$

The map gives $G\cong\mathbb{G}_m$. Such a case is not included in § 1. However, this case is much simpler. Indeed one can factorize the Equation (3.1) into the following:

$$\begin{aligned}&a\left[x+\frac{b+\varpi}{2a}y+\frac{1}{2}\left(\frac{e}{a}-\frac{2f}{\varpi}+\frac{eb}{a\varpi}\right)\right]\left[x+\frac{b-\varpi}{2a}y+\frac{1}{2}\left(\frac{e}{a}+\frac{2f}{\varpi}-\frac{eb}{a\varpi}\right)\right]\\&\quad=(bef-ce^2-af^2)(b^2-4ac)^{-1}-g.\end{aligned}$$

Define

$$f_F=2ax+(b+\varpi)y+\left(e-\frac{2af}{\varpi}+\frac{eb}{\varpi}\right)\in F[X_F]^\times$$

which induces the injection

$$f_F:\prod_{\mathfrak{p}\in\Omega_F}\mathbf{X}(\mathfrak{o}_{F_{\mathfrak{p}}})\longrightarrow \mathbb{I}_F.$$

With the above identification, one has

$$\mathbf{X}(\mathfrak{o}_F)\neq\emptyset\Leftrightarrow f_F\left[\prod_{\mathfrak{p}\in\Omega_F}\mathbf{X}(\mathfrak{o}_{F_{\mathfrak{p}}})\right]\cap[F^\times \mathrm{SO}_{\mathbb{A}}(L+u_0)]\neq\emptyset$$

by the same argument as those in Proposition 1.4. By the class field theory and [**14**, Theorem 8.1], there is a finite abelian extension K_{L+u_0}/F such that the Artin map gives

the isomorphism

$$\psi_{K_{L+u_0}/F} : \mathbb{I}_F/F^\times \mathrm{SO}_{\mathbb{A}}(L+u_0) \cong \mathrm{Gal}(K_{L+u_0}/F).$$

The above criterion can be stated as $\mathbf{X}(\mathfrak{o}_F) \neq \emptyset$, if and only if there is

$$x_A \in \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) \text{ such that } \psi_{K_{L+u_0}/F}(f_F(x_A)) = 1$$

in $\mathrm{Gal}(K_{L+u_0}/F)$. This criterion does not need any knowledge of the fixed rational point P in $X_F(F)$.

The class field theory only guarantees the existence of such abelian extension of K_{L+u_0}/F. In order to obtain the explicit condition for $\mathbf{X}(\mathfrak{o}_F) \neq \emptyset$, one needs further explicit description of K_{L+u_0}/F. There are no general theory for such explicit construction (Hilbert's 12th problem) except the cyclotomic field theory over $\mathbb{Q}$ and the complex multiplication theory of elliptic curves over imaginary quadratic fields (see [16]).

One can determine a finite set S which contains all ramification primes in K_{L+u_0}/F. Such a set S can be chosen as all primes containing ∞_F, all dyadic primes of F, all finite primes such that x_0 or u_0 is not in $L_\mathfrak{p}$, all finite primes such that $B(x_0,x_0)$ or $B(u_0,u_0)$ is not in $\mathfrak{o}_{F_\mathfrak{p}}^\times$ and all primes such that $L_\mathfrak{p}$ is not unimodular.

3.2. *Non-split cases* $-d = b^2 - 4ac \notin (F^\times)^2$

Fix the orthogonal basis $\{v_1, bv_1 - 2av_2\}$ of V. For any F-algebra A, there is a group isomorphism

$$\mathrm{SO}(V\otimes_F A) \cong \ker[(A\otimes_F E)^\times \xrightarrow{N_{E/F}} A^\times]; \sigma \mapsto \alpha + \beta\sqrt{-d}$$

such that $\sigma v_1 = \alpha v_1 + \beta(bv_1 - 2av_2)$, where $E = F(\sqrt{-d})$. This gives $G \cong R^1_{E/F}(\mathbb{G}_m)$. By this identification, one can obtain an open subgroup Ξ of $\mathbb{I}_E$ such that the map induced by inclusion

$$\mathrm{SO}_{\mathbb{A}}(V)/\mathrm{SO}(V)\mathrm{SO}_{\mathbb{A}}(L+u_0) \longrightarrow \mathbb{I}_E/E^\times\Xi$$

is injective by applying Theorem 1.10.

Let

$$f_E = 2ax + (b+\sqrt{-d})y + \left(e - \frac{2af}{\sqrt{-d}} + \frac{eb}{\sqrt{-d}}\right) \in E[X_F \times_F E]^\times$$

which induces the injection

$$f_E : \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) \longrightarrow \mathbb{I}_E.$$

By the same argument as those in Proposition 1.4, one has

$$\mathbf{X}(\mathfrak{o}_F) \neq \emptyset \Leftrightarrow f_E\left[\prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}})\right] \cap (E^\times\Xi) \neq \emptyset.$$

Let K_Ξ/E be a finite abelian extension such that

$$\psi_{K_\Xi/E} : \mathbb{I}_E/E^\times\Xi \cong \mathrm{Gal}(K_\Xi/E)$$

by the Artin map. The above criterion can be stated as $\mathbf{X}(\mathfrak{o}_F) \neq \emptyset$ if and only if there is

$$x_A \in \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}) \text{ such that } \psi_{K_\Xi/E}(f_E(x_A)) = 1 \tag{3.3}$$

in $\mathrm{Gal}(K_\Xi/E)$.

REMARK 3.4. The necessity of the above criterion does not need that Ξ is admissible. Therefore $\mathbf{X}(\mathfrak{o}_F) = \emptyset$, if there is an abelian extension K/E such that

$$\psi_{K/E}(f_E(x_A)) \neq 1 \quad \text{for any } x_A \in \prod_{\mathfrak{p}\in\Omega_F} \mathbf{X}(\mathfrak{o}_{F_\mathfrak{p}}),$$

where $\psi_{K/E}$ is the Artin map.

For determining a finite set S which contains all ramification primes in K_Ξ/F, one needs to add T_0 in Lemma 1.8 as the part of S besides the above primes of F in split case.

4. *Examples with imaginary quadratic splitting fields*

In this section, we will provide some explicit examples with imaginary quadratic splitting fields over $\mathbb{Q}$ to explain how to apply the result in § 3 to obtain the explicit conditions for $\mathbf{X}(\mathbb{Z}) \neq \emptyset$. The explicit construction of the abelian extensions of the splitting fields with the required property is crucial. For the imaginary quadratic field case, such abelian extensions can be given by the complex multiplication of elliptic curves in principle (see [**16**, Theorem 5.5]). We will keep the same notation as that in § 3.

The method described in § 3 is also the natural extension of Gauss' idea for determining the integral representation of the positive integers by a positive definite binary quadratic form. We explain this point by using one of the typical example $x^2 + dy^2$ with a positive integer d, which is studied in detail in [**4**]. Let $L = \mathbb{Z} + \mathbb{Z}\sqrt{-d}$ be an order in $E = \mathbb{Q}(\sqrt{-d})$.

PROPOSITION 4.1. *Let* $\mathbf{X}$ *be the scheme over* $\mathbb{Z}$ *defined by* $x^2 + dy^2 = n$ *for a positive integer* n, *and* K_L *be the ring class field corresponding to the order* L. *Then* $\mathbf{X}(\mathbb{Z}) \neq \emptyset$ *if and only if there is*

$$(x_p, y_p)_{p\leqslant\infty} \in \prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p) \text{ such that } \psi_{K_L/E}\left(\tilde{f}_E\left[\prod_{p\leqslant\infty}(x_p, y_p)\right]\right) = 1,$$

where $\psi_{K_L/E}$ *is the Artin map and*

$$\tilde{f}_E[(x_p, y_p)] = \begin{cases} (x_p + y_p\sqrt{-d}, x_p - y_p\sqrt{-d}) & \text{if } p \text{ splits in } E/\mathbb{Q}, \\ x_p + y_p\sqrt{-d} & \text{otherwise.} \end{cases}$$

Proof. Let p be a prime and L_p be the p-adic completion of L inside $E_p = E \otimes_{\mathbb{Q}} \mathbb{Q}_p$. Since the ring class field K_L of the order L corresponds to the open subgroup $E^\times(\prod_{p\leqslant\infty} L_p^\times)$ of $\mathbb{I}_E$ by the class field theory, one only needs to show that $\prod_{p\leqslant\infty} L_p^\times$ is a L-admissible subgroup of $\mathbb{I}_E$ with $L_\infty^\times = \mathbb{C}^\times$ by (3.3).

Since $xL_p = L_p$, for $x \in E_\mathfrak{p}^\times$ if and only if $x \in L_p^\times$, one has

$$\mathrm{SO}(L_p) = \{\alpha \in L_p^\times : N_{E_p/\mathbb{Q}_p}(\alpha) = 1\}.$$

The natural group homomorphism

$$\lambda_E : \mathrm{SO}_\mathbb{A}(V)/\mathrm{SO}(V)\mathrm{SO}_\mathbb{A}(L) \longrightarrow \mathbb{I}_E/\left(E^\times \prod_{p\leqslant\infty} L_p^\times\right)$$

is well defined.

Let $\alpha \in E^\times$ and $i \in \prod_{p\leqslant\infty} L_p^\times$ such that $\alpha i \in \mathrm{SO}_\mathbb{A}(V)$. Then

$$N_{E/\mathbb{Q}}(\alpha i) = N_{E/\mathbb{Q}}(\alpha)N_{E/\mathbb{Q}}(i) = 1.$$

Since

$$N_{E/\mathbb{Q}}(\alpha) \in \mathbb{Q} \cap \prod_{p \leqslant \infty} \mathbb{Z}_p^\times = \{\pm 1\} \quad \text{and} \quad N_{E/\mathbb{Q}}(\alpha) > 0,$$

one concludes $N_{E/\mathbb{Q}}(\alpha) = N_{E/\mathbb{Q}}(i) = 1$. This implies that

$$\alpha \in \mathrm{SO}(V) \quad \text{and} \quad i \in \mathrm{SO}_{\mathbb{A}}(L).$$

Therefore λ_E is injective and the proof is complete. □

One can recover the following classical result (see [4, Theorem 9.4]) which is obvious when $L = \mathbb{Z} + \mathbb{Z}\sqrt{-d}$ is the ring of integers of E and K_L is the Hilbert class field of E. In general, for example, if d is not square free, one needs to study the class groups of orders which are not Dedekind domains (see [4]).

COROLLARY 4.2. *Let l be an odd prime not dividing d. The equation $x^2 + dy^2 = l$ is solvable over $\mathbb{Z}$ if and only if l splits completely in K_L.*

Proof. Let $\mathbf{X}$ be the scheme over $\mathbb{Z}$ defined by the equation. By a simple computation, $\prod_{p \leqslant \infty} \mathbf{X}(\mathbb{Z}_p) \neq \emptyset$ if and only if l splits completely in E and

$$\begin{cases} \left(\frac{l}{q}\right) = 1 & \text{for each odd prime } q|d, \\ l \equiv 1 \text{ or } d \bmod 4 & \text{if } d \text{ is odd}, \\ l \equiv 1 \text{ or } d+1 \bmod 8 & \text{if } d \equiv 2 \bmod 4, \\ l \equiv 1 \bmod 4 & \text{if } \ d \equiv 4 \bmod 8, \\ l \equiv 1 \bmod 8 & \text{if } 8 \mid d. \end{cases}$$

First we show that the above local conditions can be obtained when l splits completely in K_L. Indeed, for any odd prime $q|d$, $E(\sqrt{q^*})/E$ is unramified, for all primes except the primes above q, where $q^* = (-1/q)q$. To verify if $\sqrt{q^*} \in K_L$, one only needs to see if $\psi(x)(\sqrt{q^*}) = \sqrt{q^*}$, for all $x \in \prod_{p \leqslant \infty} L_p^\times$, where ψ is the Artin map over E. This is equivalent to check if the product of the Hilbert symbols $\prod_{\mathfrak{P}} (q^*, x_{\mathfrak{P}})_{\mathfrak{P}} = 1$, where $\mathfrak{P}$ runs over all primes of E and $x = (x_{\mathfrak{P}}) \in \prod_{p \leqslant \infty} L_p^\times$. It is clear that $(q^*, x_{\mathfrak{P}})_{\mathfrak{P}} = 1$, for $\mathfrak{P} \nmid q$.

Since $x = (x_{\mathfrak{P}}) \in \prod_{p \leqslant \infty} L_p^\times$, there are $a_p, b_p \in \mathbb{Z}_p$ such that

$$\begin{cases} (x_{\mathfrak{P}}, x_{\bar{\mathfrak{P}}}) = (a_p + b_p\sqrt{-d},\ a_p - b_p\sqrt{-d}) & \text{if } p = \mathfrak{P}\bar{\mathfrak{P}} \text{ splits in } E/\mathbb{Q}, \\ x_{\mathfrak{P}} = a_p + b_p\sqrt{-d} & \text{otherwise.} \end{cases}$$

Hence

$$\prod_{\mathfrak{P}|q} (q^*, x_{\mathfrak{P}})_{\mathfrak{P}} = (q^*, a_q^2 - b_q^2 d)_q = 1$$

by [11, (1.5.3) Proposition, (7.1.4) Corollary] and $q|d$. Therefore, $\sqrt{q^*} \in K_L$. This implies that l splits in $\mathbb{Q}(\sqrt{q^*})$ and $(q^*/l) = (l/q) = 1$. This means that l is a square in $\mathbb{Z}_q$, for any odd prime $q|d$. By the Hilbert reciprocity law, one has

$$(-d, l)_2 = \prod_{\text{odd primes } q} (-d, l)_q = \prod_{q|dl} (-d, l)_q = 1.$$

One gets the local conditions for odd d and $d \equiv 2 \bmod 4$ by $(-d, l)_2 = 1$.

If $d \equiv 4 \bmod 8$, then $\mathbb{Q}(\sqrt{-1}) \subset K_L$ by the same argument as above. So $l \equiv 1 \bmod 4$.

If $8 \mid d$, then $\mathbb{Q}(\sqrt{-1}, \sqrt{2}) \subset K_L$. So $l \equiv 1 \bmod 8$.

Let $(x_p, y_p) \in \mathbf{X}(\mathbb{Z}_p)$ for $p \leqslant \infty$. Then $\tilde{f}_E[(x_p, y_p)] \in L_p^\times$ for $p \neq l$. Since l splits in $E/\mathbb{Q}$ by the local conditions and

$$(x_l + y_l\sqrt{-d})(x_l - y_l\sqrt{-d}) = l,$$

one can write $\mathfrak{L}_1$ and $\mathfrak{L}_2$ as two primes in E above l and assume $\tilde{f}_E[(x_l, y_l)]$ is a prime element at $E_{\mathfrak{L}_1}$ and a unit at $E_{\mathfrak{L}_2}$. By the class field theory, one concludes that $\mathfrak{L}_1$ splits in K_L/E if and only if

$$\psi_{K_L/E}\left(\tilde{f}_E\left[\prod_{p\leqslant\infty}(x_p, y_p)\right]\right) = 1.$$

If l splits completely in $K_L/\mathbb{Q}$, then $\mathbf{X}(\mathbb{Z}) \neq \emptyset$ by Proposition 4.1.

Conversely, choose two integral solutions (x_0, y_0) and $(x_0, -y_0)$ in $\mathbf{X}(\mathbb{Z})$. One obtains that both $\mathfrak{L}_1$ and $\mathfrak{L}_2$ split in K_L/E by the above argument. The proof is complete. □

We give one more example to explain that the method provided in the previous sections is beyond Gauss' method which only applies to the binary quadratic forms.

For any positive integer n, one can write $n = 2^s {p_1}^{e_1} \dots {p_g}^{e_g}$ and define

$$\begin{aligned} D(n) &= \{p_1, \dots, p_g\}, \\ D_1 &= \left\{p \in D(n) : \left(\frac{-1}{p}\right) = -\left(\frac{2}{p}\right) = 1\right\}, \\ D_2 &= \left\{p \in D(n) : \left(\frac{-1}{p}\right) = \left(\frac{2}{p}\right) = 1 \quad \text{and} \quad x^4 \equiv 2 \bmod p \text{ is not solvable}\right\}. \end{aligned}$$

Example 4.3. Let n be a positive integer with the above notation. Then the equation

$$x^2 + 64y^2 + 64y + 16 = n$$

is solvable over $\mathbb{Z}$ if and only if $p_j \equiv 1 \bmod 4$ for odd e_j with $1 \leqslant j \leqslant g$ and one of the following conditions holds:

(1) $s = 0$ and $n \equiv 1 \bmod 8$ and

$$\left(D_1 \neq \emptyset \text{or} \sum_{p_j \in D_2} e_j \equiv 1 \bmod 2 \quad \text{for } D_1 = \emptyset\right).$$

(2) $s = 2$ and $2^{-2}n \equiv 5 \bmod 8$.

(3) $s = 4, 5$.

Proof. Let $\mathbf{X}$ be the scheme over $\mathbb{Z}$ defined by the equation. By a simple computation, the necessary and sufficient conditions for $\prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p) \neq \emptyset$ are $p_j \equiv 1 \bmod 4$ for odd e_j with $1 \leqslant j \leqslant g$ and

$$\begin{cases} s = 0 \quad \text{and} \quad n \equiv 1 \bmod 8, \\ s = 2 \quad \text{and} \quad 2^{-2}n \equiv 5 \bmod 8, \\ s = 4, 5. \end{cases}$$

Let $E = \mathbb{Q}(\sqrt{-1})$ and $L = \mathbb{Z} + \mathbb{Z}8\sqrt{-1}$ be an order in E and $u_0 = 4\sqrt{-1}$. For any prime p, $E_p = E \otimes_{\mathbb{Q}} \mathbb{Q}_p$ and L_p is the p-adic completion of L inside E_p. Write $E_\infty = L_\infty = \mathbb{C}$. Then

$$\lambda_E[\mathrm{SO}_{\mathbb{A}}(L + u_0)] \subseteq \prod_{p\leqslant\infty} L_p^\times$$

and the induced map

$$\lambda_E : \mathrm{SO}_{\mathbb{A}}(V)/\mathrm{SO}(V)\mathrm{SO}_{\mathbb{A}}(L+u_0) \longrightarrow \mathbb{I}_E/E^\times \left(\prod_{p\leqslant\infty} L_p^\times\right)$$

is injective by the same argument in Proposition 4.1. This implies that $\prod_{p\leqslant\infty} L_p^\times$ is $(L+u_0)$-admissible. The abelian extension K_L of E corresponding to $E^\times \prod_{p\leqslant\infty} L_p^\times$ is the ring class field of L. By [4, Proposition 9.5], $K_L = E(\sqrt[4]{2})$. Then K_L/E is unramified at all places except 2 and 2 is totally ramified in $K_L/\mathbb{Q}$ and $\mathrm{Gal}(K_L/E) = \mu_4$ where μ_4 is the set of all 4th roots of unity.

Since $p \in L_2^\times$, for any odd prime p, one concludes that

$$1 = \begin{cases} \psi_{K_L/E}(p_{\mathfrak{P}}) & \text{if } p \text{ is inert in } E/\mathbb{Q} \text{ with } \mathfrak{P} \mid p, \\ \psi_{K_L/E}(p_{\mathfrak{P}_1})\psi_{K_L/E}(p_{\mathfrak{P}_2}) & \text{if } p = \mathfrak{P}\bar{\mathfrak{P}} \text{ in } E/\mathbb{Q}, \end{cases} \tag{4.4}$$

where $p_{\mathfrak{P}}$ (resp. $p_{\bar{\mathfrak{P}}}$) is in $\mathbb{I}_E$ such that its $\mathfrak{P}$ (resp. $\bar{\mathfrak{P}}$) component is p and the other components are 1 and $\psi_{K_L/E}$ is the Artin symbol.

Define

$$\tilde{f}_E[(x_p, y_p)] = \begin{cases} (x_p + 4(2y_p+1)\sqrt{-1}, x_p - 4(2y_p+1)\sqrt{-1}) & \text{if } p \text{ splits in } E/\mathbb{Q}, \\ x_p + 4(2y_p+1)\sqrt{-1} & \text{otherwise,} \end{cases}$$

for any $(x_p, y_p) \in \mathbf{X}(\mathbb{Z}_p)$.

If $(p, 2n) = 1$, then

$$\tilde{f}_E[(x_p, y_p)] \in L_p^\times \quad \text{and} \quad \psi_{K_L/E}(\tilde{f}_E[(x_p, y_p)]) = 1,$$

where $\tilde{f}_E[(x_p, y_p)]$ is also regarded as an element in $\mathbb{I}_E$ such that the components above p are given by the value of $\tilde{f}_E[(x_p, y_p)]$ and 1 otherwise.

If p is inert in $E/\mathbb{Q}$ with $p \in D(n)$, then $\psi_{K_L/E}(\tilde{f}_E[(x_p, y_p)]) = 1$ by (4.4).

If p splits in $E/\mathbb{Q}$ with $p \in D(n)$, one has

$$\begin{cases} \psi_{K_L/E}(p_{\mathfrak{P}}) = \psi_{K_L/E}(p_{\bar{\mathfrak{P}}}) = -1 & \text{for } p \in D_2, \\ \psi_{K_L/E}(p_{\mathfrak{P}}) = -\psi_{K_L/E}(p_{\bar{\mathfrak{P}}}) = \pm\sqrt{-1} & \text{for } p \in D_1 \end{cases}$$

by the definition of D_1, D_2 and (4.4). Since

$$(x_p + 4(2y_p+1)\sqrt{-1})(x_p - 4(2y_p+1)\sqrt{-1}) = n,$$

one has

$$\psi_{K_L/E}(\tilde{f}_E[(x_p, y_p)]) = \begin{cases} (-1)^e & \text{if } p \in D_2, \\ (-1)^a(\pm\sqrt{-1})^e & \text{if } p \in D_1, \\ 1 & \text{otherwise,} \end{cases} \tag{4.5}$$

where $a = \mathrm{ord}_{\mathfrak{P}}(x_p + 4(2y_p+1)\sqrt{-1})$ and e is the exponent of p inside n. By the Hensel's lemma, there are two local solutions $(x_p, y_p) \in \mathbf{X}(\mathbb{Z}_p)$ such that $a = 0$ and 1, respectively. Therefore, one can rewrite (4.5), for $p \in D_1$, as

$$\psi_{K_L/E}(\tilde{f}_E[(x_p, y_p)]) = \begin{cases} \pm\sqrt{-1} & e \text{ is odd}, \\ \pm 1 & e \text{ is even}, \end{cases} \tag{4.6}$$

where the sign can be chosen as one wishes.

Suppose $s = 4$ or 5. Since $x^2 + (2y+1)^2 = 2^{-4}n$ is solvable over $\mathbb{Z}$ by the above local conditions, one concludes that $x^2 + 16(2y+1)^2 = n$ is solvable over $\mathbb{Z}$.

Suppose $s = 2$. Then the local solution $(x_2, y_2) \in \mathbf{X}(\mathbb{Z}_2)$ satisfies $x_2 \equiv 2 \bmod 4$. Let v be the unique place of E over 2. Since

$$(2, x_2 + 4(2y_2 + 1)\sqrt{-1})_v = (2, 2)_v \cdot (2, 2^{-1}x_2 + 2(2y_2 + 1)\sqrt{-1})_v = (2, 5)_2 = -1$$

by [**11**, (1.5.3) Proposition and (7.1.4) Corollary], this implies that

$$\psi_{K_L/E}(\tilde{f}_E[(x_2, y_2)]) = \pm\sqrt{-1}.$$

Since $2^{-2}n \equiv 5 \bmod 8$, the number of primes p_j in D_1 with odd e_j is odd. Therefore, there are an even number of local primes p such that

$$\psi_{K_L/E}(\tilde{f}_E[(x_p, y_p)]) = \pm\sqrt{-1}$$

by the above computation and (4.6). By choosing the sign properly, one concludes that there is

$$(x_p, y_p)_{p\leqslant\infty} \in \prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p) \text{ such that } \psi_{K_L/E}\left(\tilde{f}_E\left[\prod_{p\leqslant\infty}(x_p, y_p)\right]\right) = 1.$$

Therefore, $\mathbf{X}(\mathbb{Z}) \neq \emptyset$ by (3.3).

Suppose $s = 0$. Then the local solution $(x_2, y_2) \in \mathbf{X}(\mathbb{Z}_2)$ satisfies $x_2 \equiv 1 \bmod 2$. Let v be the unique place of E over 2. Since

$$\tilde{f}_E[(x_2, y_2)] = x_2 + 4(2y_2 + 1)\sqrt{-1} \equiv x_2 + 4\sqrt{-1} \equiv 1 + 4\sqrt{-1} \bmod L_2^{\times},$$

and $(1 + 4\sqrt{-1}) \in E^{\times}$ with $(1 + 4\sqrt{-1})(1 - 4\sqrt{-1}) = 17$, one concludes that $\psi_{K_L/E}(\tilde{f}_E[(x_2, y_2)])$ is equal to the Frobenius of a prime above 17 in $\mathrm{Gal}(K_L/E)$ which is -1.

If $D_1 \neq \emptyset$, one has that either all e_j for $p_j \in D_1$ are even or the number of primes p_j in D_1 with odd e_j is even by using $n \equiv 1 \bmod 8$. By (4.6) and choosing the sign properly, there is

$$(x_p, y_p)_{p\leqslant\infty} \in \prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p) \text{ such that } \psi_{K_L/E}\left(\tilde{f}_E\left[\prod_{p\leqslant\infty}(x_p, y_p)\right]\right) = 1.$$

Therefore, $\mathbf{X}(\mathbb{Z}) \neq \emptyset$ by (3.3).

If $D_1 = \emptyset$, then

$$\psi_{K_L/E}\left(\tilde{f}_E\left[\prod_{p\leqslant\infty}(x_p, y_p)\right]\right) = (-1)^{1+\sum_{p_j\in D_2} e_j}$$

by (4.5), for any $(x_p, y_p)_{p\leqslant\infty} \in \prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p)$. By (3.3), one concludes

$$\mathbf{X}(\mathbb{Z}) \neq \emptyset \Leftrightarrow \sum_{p_j\in D_2} e_j \equiv 1 \bmod 2$$

and the proof is complete. □

5. *Examples with real quadratic splitting fields*

Gauss' method cannot be applied to integral representations of indefinite binary quadratic forms in general either because the norms of the elements of real quadratic fields are not always positive and one cannot decide the sign effectively. The typical example is to determine exactly when the negative Pell equation $x^2 - dy^2 = -1$ is solvable over $\mathbb{Z}$ for a positive integer d.

In order to apply the method in § 3, for example, to determine all integers represented by $x^2 - dy^2$, for a positive integer d explicitly, one can take the order $L = \mathbb{Z} + \mathbb{Z}\sqrt{d}$ inside $E = \mathbb{Q}(\sqrt{d})$ and needs to determine the abelian extension of E corresponding to an L-admissible subgroup of $\mathbb{I}_E$ explicitly. There is no general theory for obtaining such abelian extensions for real quadratic fields but *ad hoc* method. We will keep the same notation as that in § 3.

With some extra conditions, one still has the analogous result to Proposition 4.1 for real quadratic fields.

PROPOSITION 5.1. *Let* $\mathbf{X}$ *be the scheme defined by* $x^2 - dy^2 = n$, *for some integer* n *and* K_L *be the ring class field corresponding to the order* L. *Suppose one of the following conditions holds*:

(1) *the equation* $x^2 - dy^2 = -1$ *has an integer solution.*
(2) *the equation* $x^2 - dy^2 = -1$ *has no local integral solutions at a prime.*

Then $\mathbf{X}(\mathbb{Z}) \neq \emptyset$ *if and only if there is*

$$(x_p, y_p)_{p\leqslant\infty} \in \prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p) \text{ such that } \psi_{K_L/E}\left(\tilde{f}_E\left[\prod_{p\leqslant\infty}(x_p, y_p)\right]\right) = 1,$$

where $\psi_{K_L/E}$ *is the Artin map and*

$$\tilde{f}_E[(x_p, y_p)] = \begin{cases} (x_p + y_p\sqrt{d}, x_p - y_p\sqrt{d}) & \text{if } p \text{ splits in } E/\mathbb{Q}, \\ x_p + y_p\sqrt{d} & \text{otherwise.} \end{cases}$$

Proof. The same argument as that in Proposition 4.1 with the crucial step to show that λ_E is injective is available except $N_{E/\mathbb{Q}}(\alpha) > 0$. Suppose $N_{E/\mathbb{Q}}(\alpha) = -1$. Then $N_{E/\mathbb{Q}}(i) = -1$. This contradicts Condition (2). If Condition (1) holds, there are $x_0, y_0 \in \mathbb{Z}$ such that $x_0^2 - dy_0^2 = -1$. One can replace α by $\alpha(x_0 - y_0\sqrt{d})$ and i by $i(x_0 - y_0\sqrt{d})^{-1}$. The proof is complete. □

When d is a prime, the condition (1) or (2) in Proposition 5.1 will be satisfied. For $d = 2l$ with a prime l, the condition (2) in Proposition 5.1 holds at prime 2, if $l \equiv 3 \bmod 4$ and the condition (1) in Proposition 5.1 holds if $l \equiv 5 \bmod 8$ by [5]. Such cases can also be treated by using the Gauss' method.

However, the Gauss' method only uses the ring class fields to detect which ideals are principal. Now we consider the case that one has to use the abelian extensions beyond the ring class fields. Let l be a prime with $l \equiv 1 \bmod 8$. Fix an integral solution (x_0, y_0, z_0) of the equation $x^2 - 2ly^2 = 2z^2$ such that $x_0 > 0$ and $(x_0, y_0, z_0) = 1$ by the Hasse principle. Let $\Theta = E(\sqrt{x_0 - y_0\sqrt{2l}})$. Then Θ is totally real and Θ/E is unramified over all primes except the prime above 2 and 2 is totally ramified in $\Theta/\mathbb{Q}$. This Θ will play role for solving the equation $x^2 - 2ly^2 = n$ over $\mathbb{Z}$.

LEMMA 5.2. *Let* l *be a prime with* $l \equiv 1 \bmod 8$ *satisfying* $2l = r^2 + s^2$ *for two integers* r *and* s *with* $s \equiv \pm 3 \bmod 8$. *If* x_2 *and* y_2 *in* $\mathbb{Q}_2$ *satisfy* $x_2^2 - 2ly_2^2 = -1$, *then the Hilbert symbol*

$$(x_2 + y_2\sqrt{2l}, x_0 - y_0\sqrt{2l})_v = -1,$$

where v *is the unique prime in* E *above* 2 *and* (x_0, y_0) *is given as above.*

Proof. For any $\xi \in E_v^\times$ with $N_{L_v/\mathbb{Q}_2}(\xi) = 1$, there exists $\beta \in E_v^\times$ such that $\xi = \sigma(\beta)\beta^{-1}$ by Hilbert 90, where σ is the non-trivial element in $\mathrm{Gal}(E_v/\mathbb{Q}_2)$. Then

$$\begin{aligned}(\xi, x_0 - y_0\sqrt{2l})_v &= (\sigma(\beta)\beta^{-1}, x_0 - y_0\sqrt{2l})_v \\ &= (N_{E_v/\mathbb{Q}_2}(\beta), x_0 - y_0\sqrt{2l})_v = (N_{E_v/\mathbb{Q}_2}(\beta), 2)_2\end{aligned}$$

by [11, (1.5.3) Proposition and (7.1.4) Corollary]. Since $l \equiv 1 \bmod 8$ and $E_v = \mathbb{Q}_2(\sqrt{2l})$, one has

$$(\xi, x_0 - y_0\sqrt{2l})_v = (N_{E_v/\mathbb{Q}_2}(\beta), 2)_2 = (N_{E_v/\mathbb{Q}_2}(\beta), 2l)_2 = 1.$$

Therefore,

$$\begin{aligned}(x_2 + y_2\sqrt{2l}, x_0 - y_0\sqrt{2l})_v &= ((r - \sqrt{2l})s^{-1}, x_0 - y_0\sqrt{2l})_v \\ &= (r - \sqrt{2l}, x_0 - y_0\sqrt{2l})_v(s, x_0 - y_0\sqrt{2l})_v \\ &= (r - \sqrt{2l}, x_0 - y_0\sqrt{2l})_v(s, 2z_0^2)_2 = -(r - \sqrt{2l}, x_0 - y_0\sqrt{2l})_v.\end{aligned}$$

By the Hilbert reciprocity law, one has

$$(r - \sqrt{2l}, x_0 - y_0\sqrt{2l})_v = \prod_{\mathfrak{p}\neq v}(r - \sqrt{2l}, x_0 - y_0\sqrt{2l})_{\mathfrak{p}}.$$

Since

$$(r - \sqrt{2l})(r + \sqrt{2l}) = -s^2 \quad \text{and} \quad (x_0 - y_0\sqrt{2l})(x_0 + y_0\sqrt{2l}) = 2z_0^2$$

with $(r, s) = 1$ and $(x_0, y_0, z_0) = 1$, respectively, one has

$$\operatorname{ord}_{\mathfrak{p}}(r - \sqrt{2l}) \equiv \operatorname{ord}_{\mathfrak{p}}(x_0 - y_0\sqrt{2l}) \equiv 0 \bmod 2,$$

for $\mathfrak{p} \neq v$ and $\mathfrak{p} < \infty_E$. Since $x_0 - y_0\sqrt{2l} > 0$ over $\mathfrak{p} \in \infty_E$, one obtains that

$$(r - \sqrt{2l}, x_0 - y_0\sqrt{2l})_{\mathfrak{p}} = 1,$$

for all $\mathfrak{p} \neq v$. One concludes that $(x_2 + y_2\sqrt{2l}, x_0 - y_0\sqrt{2l})_v = -1$. □

The immediate corollary is the following result which was first proved in [6] by using the continued fractional method and was reproved in [15] by using the reciprocity law in a quite complicated way.

COROLLARY 5.3 (Epstein–Rédei). *If l is a prime with $l \equiv 1 \bmod 8$ satisfying $2l = r^2 + s^2$ for two integers r and s with $s \equiv \pm 3 \bmod 8$, then the equation $x^2 - 2ly^2 = -1$ is not solvable over $\mathbb{Z}$.*

Proof. Let $\mathbf{X}$ be the scheme defined by $x^2 - 2ly^2 = -1$. Then

$$\psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)_{p\leqslant\infty}]) = (x_2 + y_2\sqrt{2l}, x_0 - y_0\sqrt{2l})_v = -1,$$

for any $\prod_{p\leqslant\infty}(x_p, y_p) \in \prod_{p\leqslant\infty}\mathbf{X}(\mathbb{Z}_p)$ by Lemma 5.2. The result follows from Remark 3.4. □

It is natural to ask how to decide the solvability of the above equation if one replaces -1 by an integer n.

PROPOSITION 5.4. *Let l be a prime as in Lemma 5.2 and $\mathbf{X}$ be the scheme over $\mathbb{Z}$ defined by $x^2 - 2ly^2 = n$, for some integer n. Then $\mathbf{X}(\mathbb{Z}) \neq \emptyset$ if and only if there is $(x_p, y_p)_{p\leqslant\infty} \in \prod_{p\leqslant\infty}\mathbf{X}(\mathbb{Z}_p)$ such that*

$$\psi_{H/E}(\tilde{f}_E[(x_p, y_p)_{p\leqslant\infty}]) = 1 \quad \text{and} \quad \psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)_{p\leqslant\infty}]) = 1,$$

where H is the Hilbert class field of E, Θ is the quadratic extension of E defined as above, both $\psi_{H/E}$ and $\psi_{\Theta/E}$ are Artin maps and

$$\tilde{f}_E[(x_p, y_p)] = \begin{cases}(x_p + y_p\sqrt{2l}, x_p - y_p\sqrt{2l}) & \text{if } p \text{ splits in } E/\mathbb{Q}, \\ x_p + y_p\sqrt{2l} & \text{otherwise.}\end{cases}$$

Proof. Let $L = \mathbb{Z} + \mathbb{Z}\sqrt{2l}$ be the maximal order of E. For any prime p, $E_p = E \otimes_{\mathbb{Q}} \mathbb{Q}_p$ and L_p is the p-adic completion of L inside E_p. Write $E_\infty = L_\infty = \mathbb{R}$.

Let v be the unique prime of E above 2. By the computation in Lemma 5.2, one has $(\xi, x_0 - y_0\sqrt{2l})_v = 1$, for any $\xi \in E_v^\times$ with $N_{E_v/\mathbb{Q}_2}(\xi) = 1$. This implies that

$$\lambda_E(\mathrm{SO}(L_2)) \subseteq N_{\Theta_{\mathfrak{V}}/E_v}(\Theta_{\mathfrak{V}}^\times),$$

where $\mathfrak{V}$ is the unique prime of Θ above v. Since Θ/E is unramified over all primes except v, one concludes that $\lambda_E(\mathrm{SO}_{\mathbb{A}}(L)) \subseteq N_{\Theta/E}(\mathbb{I}_\Theta)$. Therefore the natural group homomorphism

$$\lambda_E : \mathrm{SO}_{\mathbb{A}}(V)/\mathrm{SO}(V)\mathrm{SO}_{\mathbb{A}}(L) \longrightarrow [\mathbb{I}_E/(E^\times N_{\Theta/E}(\mathbb{I}_\Theta))] \times \left[\mathbb{I}_E/\left(E^\times \prod_{p\leqslant\infty} L_p^\times\right)\right]$$

is well defined.

Let $u \in \ker\lambda_E$. Then there are $\alpha \in E^\times$ and $i \in \prod_{p\leqslant\infty} L_p^\times$ with $\lambda_E(u) = \alpha i$. Therefore,

$$N_{E/\mathbb{Q}}(\alpha) = N_{E/\mathbb{Q}}(i)^{-1} \in \mathbb{Q}^\times \cap \left(\prod_{p\leqslant\infty} \mathbb{Z}_p^\times\right) = \{\pm 1\}.$$

Suppose $N_{E/\mathbb{Q}}(\alpha) \neq 1$. Then $N_{E/\mathbb{Q}}(\alpha) = N_{E/\mathbb{Q}}(i) = -1$. Write $i = (i_{\mathfrak{p}})_{\mathfrak{p}} \in \mathbb{I}_E$. Since Θ/E is unramified over all primes of E except v, one has $\psi_{\Theta/E}(i_{\mathfrak{p}})$ is trivial, for all primes $\mathfrak{p} \neq v$, where $i_{\mathfrak{p}}$ is regarded as an idele whose $\mathfrak{p}$-component is $i_{\mathfrak{p}}$ and 1 otherwise. Since $N_{E/\mathbb{Q}}(i_v) = N_{E_v/\mathbb{Q}_2}(i_v) = -1$, one gets

$$\psi_{\Theta/E}(\alpha i) = \psi_{\Theta/E}(i) = \psi_{\Theta/E}(i_v) = -1$$

by Lemma 5.2. This contradicts $u \in \ker\lambda_E$.

Therefore, $N_{E/\mathbb{Q}}(\alpha) = N_{E/\mathbb{Q}}(i) = 1$. This implies that $\alpha \in \mathrm{SO}(V)$ and $i \in \mathrm{SO}_{\mathbb{A}}(L)$. One concludes that λ_E is injective. The result follows from Corollary 1.6. □

Finally, we will use Proposition 5.4 to give an explicit example. For any integer n, one can write $n = (-1)^{s_0} 2^{s_1} 17^{s_2} {p_1}^{e_1} \dots {p_g}^{e_g}$ and $D(n) = \{p_1, \dots, p_g\}$. Decompose $D(n)$ into the disjoint union of the following subsets:

$$D_1 = \left\{p \in D(n) : \left(\frac{2}{p}\right) = \left(\frac{17}{p}\right) = -1\right\} \quad \text{and} \quad D_2 = \left\{p \in D(n) : \left(\frac{34}{p}\right) = -1\right\},$$

$$D_3 = \left\{p \in D(n) : \left(\frac{2}{p}\right) = \left(\frac{17}{p}\right) = 1 \quad \text{and} \quad x^4 - 12x^2 + 2 \equiv 0 \bmod p \text{ is solvable}\right\},$$

$$D_4 = \left\{p \in D(n) : \left(\frac{2}{p}\right) = \left(\frac{17}{p}\right) = 1 \quad \text{and} \quad x^4 - 12x^2 + 2 \equiv 0 \bmod p \text{ is not solvable}\right\}.$$

Let

$$n_1 = (-1)^{s_0} \prod_{p_i \in D(n)\setminus D_2} p_i^{e_i}.$$

Example 5.5. Let n be an integer with the above notation. Then the equation

$$x^2 - 34y^2 = n$$

is solvable over $\mathbb{Z}$ if and only if

(1) $n_1 \equiv \pm 1 \bmod 8$, $(n_1/17) = 1$ and $(34/p_i) = 1$ for odd e_i

(2) $D_1 \neq \emptyset$; or

$$\sum_{p_i \in D_4} e_i \equiv \begin{cases} s_2 \bmod 2 & \text{if } n_1 \equiv 1, -9 \bmod 16, \\ s_2 + 1 \bmod 2 & \text{if } n_1 \equiv -1, 9 \bmod 16, \end{cases}$$

for $D_1 = \emptyset$.

Proof. Let $\mathbf{X}$ be the scheme over $\mathbb{Z}$ defined by the equation. By [12] for odd primes p and [9] for $p = 2$, one has the condition (1) is equivalent to $\prod_{p \leqslant \infty} \mathbf{X}(\mathbb{Z}_p) \neq \emptyset$.

Since $34 = 5^2 + 3^2$, Proposition 5.4 can be applied. Moreover, the Hilbert class field H of $E = \mathbb{Q}(\sqrt{34})$ is $\mathbb{Q}(\sqrt{2}, \sqrt{17})$. Since the equation $x^2 - 34y^2 = 2$ has an integral solution for $x = 6$ and $y = 1$, one can choose $\Theta = E(\sqrt{6 - \sqrt{34}})$. For simplicity, we will denote both $\mathrm{Gal}(\Theta/E)$ and $\mathrm{Gal}(H/E)$ as $\mu_2 = \{\pm 1\}$.

Define

$$\tilde{f}_E[(x_p, y_p)] = \begin{cases} (x_p + y_p\sqrt{34}, x_p - y_p\sqrt{34}) & \text{if } p \text{ splits in } E/\mathbb{Q}, \\ x_p + y_p\sqrt{34} & \text{otherwise,} \end{cases}$$

for any $(x_p, y_p) \in \mathbf{X}(\mathbb{Z}_p)$.

Then

$$\psi_{H/E}(\tilde{f}_E[(x_p, y_p)]) = \begin{cases} (-1)^{e_i} & \text{if } p \in D_1, \\ 1 & \text{otherwise.} \end{cases}$$

Since $n_1 \equiv \pm 1 \bmod 8$ by the local condition (1), one always has $\sum_{p_i \in D_1} e_i \equiv 0 \bmod 2$. This implies that

$$\psi_{H/E}(\tilde{f}_E[(x_p, y_p)_{p \leqslant \infty}]) = 1 \quad \text{for any} \quad (x_p, y_p)_{p \leqslant \infty} \in \prod_{p \leqslant \infty} \mathbf{X}(\mathbb{Z}_p).$$

Next we compute $\psi_{\Theta/E}$. It is clear that $\psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)]) = 1$ for $p \nmid 34n$.

If $p \in D_1$, then $p = \mathfrak{P}\bar{\mathfrak{P}}$ over E. Moreover, one of $\mathfrak{P}$ and $\bar{\mathfrak{P}}$ splits in Θ/E and the other is inert in Θ/E. Without loss of generality, we assume that $\mathfrak{P}$ is inert in Θ/E. Then

$$\psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)]) = (-1)^a \quad \text{with } a = ord_{\mathfrak{P}}(x_p + y_p\sqrt{34}).$$

If $p \in D_3$, then p splits completely in $\Theta/\mathbb{Q}$ and $\psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)]) = 1$.

If $p \in D_4$, then $p = \mathfrak{P}\bar{\mathfrak{P}}$ over E and both $\mathfrak{P}$ and $\bar{\mathfrak{P}}$ are inert in Θ/E. One gets

$$\psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)]) = (-1)^e,$$

where e is the exponent of p inside n.

If $p \in D_2$, the exponent e of p inside n satisfies $e \equiv 0 \bmod 2$ by the condition (1) and p is inert in $E/\mathbb{Q}$. Moreover, the prime $\mathfrak{P}$ above p in E splits in Θ/E if and only if $6 - \sqrt{34}$ is a square in $E_{\mathfrak{P}}$. By the Hensel's lemma, this is equivalent to $(6 - \sqrt{34}, p)_{\mathfrak{P}} = (2, p)_p = 1$. Since $ord_{\mathfrak{P}}(x_p + y_p\sqrt{34}) = \frac{1}{2}\mathrm{ord}_p(n)$, one has

$$\psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)]) = \left(\frac{2}{p}\right)^{(1/2)e}.$$

We summarize

$$\psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)]) = \begin{cases} 1 & \text{if } (p, 34n) = 1 \text{ or } p \in D_3, \\ (-1)^{s_2} & \text{if } p = 17, \\ (-1)^a & \text{if } p \in D_1, \\ \left(\frac{2}{p}\right)^{(1/2)e} & \text{if } p \in D_2, \\ (-1)^e & \text{if } p \in D_4, \end{cases}$$

for any $(x_p, y_p) \in \mathbf{X}(\mathbb{Z}_p)$.

When $D_1 \neq \emptyset$, there are two local solutions $(x_p, y_p) \in \mathbf{X}(\mathbb{Z}_p)$ such that $a = 0$ and 1, respectively, for any $p \in D_1$ by the Hensel's Lemma. This implies that there is $(x_p, y_p)_{p\leqslant\infty} \in \prod_{p\leqslant\infty} \mathbf{X}(\mathbb{Z}_p)$ such that

$$\psi_{\Theta/E}(\tilde{f}_E[(x_p, y_p)_{p\leqslant\infty}]) = 1.$$

When $D_1 = \emptyset$, one needs further to compute $\psi_{\Theta/E}(\tilde{f}_E[(x_2, y_2)])$. Since there is $\delta \in \mathbb{Z}_2^\times$ with

$$\delta \equiv \begin{cases} \pm 1 \bmod 8 & \text{if } n_1 \equiv \pm 1 \bmod 16, \\ \pm 3 \bmod 8 & \text{if } n_1 \equiv \pm 9 \bmod 16 \end{cases}$$

such that

$$\delta^2 = \begin{cases} 17^{s_2} n_1 & \text{if } n_1 \equiv 1,\ 9 \bmod 16, \\ -17^{s_2} n_1 & \text{if } n_1 \equiv -1,\ -9 \bmod 16 \end{cases}$$

by the Hensel's lemma, one has

$$N_{E_v/\mathbb{Q}_2}\left(\beta^{s_1}\delta \prod_{p_i\in D_2} p_i^{(1/2)e_i}\right) = \begin{cases} n & \text{if } n_1 \equiv 1,\ 9 \bmod 16, \\ -n & \text{if } n_1 \equiv -1, -9 \bmod 16, \end{cases}$$

where v is the unique prime of E above 2 and $\beta = 6 - \sqrt{34}$. For any $(x_2, y_2) \in \mathbf{X}(\mathbb{Z}_2)$, one has

$$\psi_{\Theta/E}(\tilde{f}_E[(x_2, y_2)]) = \begin{cases} (\beta^{s_1}\delta \prod_{p_i\in D_2} p_i^{(1/2)e_i}, \beta)_v & \text{if } n_1 \equiv 1, 9 \bmod 16, \\ -(\beta^{s_1}\delta \prod_{p_i\in D_2} p_i^{(1/2)e_i}, \beta)_v & \text{if } n_1 \equiv -1, -9 \bmod 16 \end{cases}$$

by the computation in Lemma 5.2. Since

$$\begin{aligned} \left(\beta^{s_1}\delta \prod_{p_i\in D_2} p_i^{(1/2)e_i}, \beta\right)_v &= (\beta, \beta)_v^{s_1} \left(\delta \prod_{p_i\in D_2} p_i^{(1/2)e_i}, \beta\right)_v \\ &= (\beta, -1)_v^{s_1} \left(\delta \prod_{p_i\in D_2} p_i^{(1/2)e_i}, 2\right)_2 = (2, -1)_2^{s_1} \left(\delta \prod_{p_i\in D_2} p_i^{(1/2)e_i}, 2\right)_2 \\ &= (\delta, 2)_2 \prod_{p_i\in D_2} \left(\frac{2}{p_i}\right)^{(1/2)e_i} \\ &= \begin{cases} \prod_{p_i\in D_2} \left(\frac{2}{p_i}\right)^{(1/2)e_i} & \text{if } n_1 \equiv \pm 1 \bmod 16, \\ -\prod_{p_i\in D_2} \left(\frac{2}{p_i}\right)^{(1/2)e_i} & \text{if } n_1 \equiv \pm 9 \bmod 16, \end{cases} \end{aligned}$$

one concludes that

$$\psi_{\Theta/E}(\tilde{f}_E[(x_2, y_2)]) = \begin{cases} \prod_{p_i\in D_2} \left(\frac{2}{p_i}\right)^{(1/2)e_i} & \text{if } n_1 \equiv 1, -9 \bmod 16, \\ -\prod_{p_i\in D_2} \left(\frac{2}{p_i}\right)^{(1/2)e_i} & \text{if } n_1 \equiv -1, 9 \bmod 16, \end{cases}$$

for any $(x_2, y_2) \in \mathbf{X}(\mathbb{Z}_2)$. By Proposition 5.4, $\mathbf{X}(\mathbb{Z}) \neq \emptyset$ if and only if

$$\prod_{p_i\in D_4} (-1)^{e_i} \prod_{p_i\in D_2} \left(\frac{2}{p_i}\right)^{(1/2)e_i} = \begin{cases} (-1)^{s_2} \prod_{p_i\in D_2} \left(\frac{2}{p_i}\right)^{(1/2)e_i} & \text{if } n_1 \equiv 1, -9 \bmod 16, \\ (-1)^{s_2+1} \prod_{p_i\in D_2} \left(\frac{2}{p_i}\right)^{(1/2)e_i} & \text{if } n_1 \equiv -1, 9 \bmod 16. \end{cases}$$

The proof is complete. □

6. *Examples for higher-dimensional tori*

In this section, we further explain how effective the method in § 1 is for high dimension tori by providing the explicit examples. Fix an integral solution (x_0, y_0, z_0) of the equation $x^2 - 2ly^2 = 2z^2$ for $l \equiv 1 \bmod 8$ such that $x_0 > 0$ and $(x_0, y_0, z_0) = 1$. Let

$$L = \mathbb{Q}(\sqrt{2l}) \quad \text{and} \quad \Theta = L(\sqrt{x_0 - y_0\sqrt{2l}}).$$

Then Θ is totally real and Θ/L is unramified over all primes except the prime above 2 and 2 is totally ramified in $\Theta/\mathbb{Q}$.

PROPOSITION 6.1. *Let l be a prime with $l \equiv 1 \bmod 8$ satisfying $2l = r^2 + s^2$ for two integers r and s with $s \equiv \pm 3 \bmod 8$ and Θ and L be as above. Suppose E is a field containing L and $\mathbf{X}$ is the scheme over $\mathbb{Z}$ defined by the equation*

$$f(x_1, \ldots, x_k) = N_{E/\mathbb{Q}}(x_1 e_1 + \ldots + x_s e_k) = n,$$

where $\{e_1, \ldots, e_k\}$ is a basis of $\mathfrak{o}_E$ over $\mathbb{Z}$ and n is an integer. Then $\mathbf{X}(\mathbb{Z}) \neq \emptyset$ if and only if $X_{\mathbb{Q}}(\mathbb{Q}) \neq \emptyset$ and there is

$$(x_{1p}, \ldots, x_{kp})_{p \leqslant \infty} \in \prod_{p \leqslant \infty} \mathbf{X}(\mathbb{Z}_p)$$

such that both

$$\psi_{H/E}[(x_{1p}e_1 + \ldots + x_{kp}e_k)_{p\leqslant\infty}] \quad \text{and} \quad \psi_{\Theta/L}[(N_{E/L}(x_{1p}e_1 + \ldots + x_{kp}e_k))_{p\leqslant\infty}]$$

are trivial, where H is the Hilbert class field of E and both $\psi_{H/E}$ and $\psi_{\Theta/L}$ are the Artin maps.

Proof. Since $G = R^1_{E/\mathbb{Q}}(\mathbb{G}_m)$, one has

$$\mathrm{Stab}_{\mathbb{A}}(\mathbf{X}) = \left\{ (x_p) \in \prod_{p \leqslant \infty} \mathfrak{o}_{E_p}^{\times} : N_{E/\mathbb{Q}}((x_p)) = 1 \in \mathbb{I}_{\mathbb{Q}} \right\}.$$

By the computation in Lemma 5.2, one has $(\xi, x_0 - y_0\sqrt{2l})_v = 1$, for any $\xi \in L_v^{\times}$ with $N_{L_v/\mathbb{Q}_2}(\xi) = 1$ where v is the unique prime of L above 2. This implies that

$$N_{E/L}(\mathrm{Stab}(\mathbf{X}(\mathbb{Z}_2)) \subseteq N_{\Theta_{\mathfrak{V}}/L_v}(\Theta_{\mathfrak{V}}^{\times}),$$

where $\mathfrak{V}$ is the unique prime of Θ above v. Since Θ/L is unramified over all primes except v, one concludes that

$$N_{E/L}(\mathrm{Stab}_{\mathbb{A}}(\mathbf{X})) \subseteq N_{\Theta/L}(\mathbb{I}_{\Theta}),$$

where $\mathbb{I}_{\Theta}$ is the idele group of Θ. Then the group homomorphism

$$f_E : G(\mathbb{A}_{\mathbb{Q}})/G(\mathbb{Q})\mathrm{Stab}_{\mathbb{A}}(\mathbf{X}) \longrightarrow [\mathbb{I}_L/(L^{\times}N_{\Theta/L}(\mathbb{I}_{\Theta}))] \times \left[\mathbb{I}_E \Big/ \left(E^{\times} \prod_{p \leqslant \infty} \mathfrak{o}_{E_p}^{\times} \right) \right],$$
$$(x_p)_{p\leqslant\infty} \mapsto (N_{E/L}(x_p), (x_p))_{p\leqslant\infty}$$

is well defined. By Lemma 5.2 and the same argument in Proposition 5.4, one obtains that f_E is injective and the result follows. □

For any integer n, one can write

$$n = (-1)^{s_0} 2^{s_1} 17^{s_2} {p_1}^{e_1} \ldots {p_g}^{e_g} \quad \text{and} \quad D(n) = \{p_1, \ldots, p_g\}.$$

Decompose $D(n)$ into the disjoint union of the following subsets:

$$D_1 = \left\{ p \in D(n) : \left(\frac{2}{p}\right) = \left(\frac{17}{p}\right) = -1 \right\} \quad \text{and} \quad D_2 = \left\{ p \in D(n) : \left(\frac{34}{p}\right) = -1 \right\},$$

$$D_3 = \left\{ p \in D(n) : \left(\frac{2}{p}\right) = \left(\frac{17}{p}\right) = 1 \quad \text{and} \quad x^4 - 12x^2 + 2 \equiv 0 \bmod p \text{ is solvable} \right\},$$

$$D_4 = \left\{ p \in D(n) : \left(\frac{2}{p}\right) = \left(\frac{17}{p}\right) = 1 \quad \text{and} \quad x^4 - 12x^2 + 2 \equiv 0 \bmod p \text{ is not solvable} \right\}.$$

Let

$$n_1 = (-1)^{s_0} \prod_{p_i \in D(n) \setminus D_2} p_i^{e_i}.$$

One can have one more explicit example.

EXAMPLE 6.2. Let n be an integer with the above notation and $E = \mathbb{Q}(\sqrt{5}, \sqrt{34})$. Then $n \in N_{E/\mathbb{Q}}(\mathfrak{o}_E)$ if and only if

(1) $s_1 \equiv s_2 \equiv 0 \bmod 2$, $n_1 \equiv \pm 1 \bmod 8$, $(n_1/17) = 1$ and $(34/p_i) = (5//p_i) = 1$ for odd e_i;
(2) $D_1 \neq \emptyset$; or

$$\sum_{p_i \in D_4} e_i \equiv \begin{cases} 0 \bmod 2 & \text{if } n_1 \equiv 1, -9 \bmod 16, \\ 1 \bmod 2 & \text{if } n_1 \equiv -1, 9 \bmod 16, \end{cases}$$

for $D_1 = \emptyset$.

Proof. We apply Proposition 6.1 by taking $E = \mathbb{Q}(\sqrt{5}, \sqrt{34})$, $L = \mathbb{Q}(\sqrt{34})$ and $\Theta = L(\sqrt{6 - \sqrt{34}})$. The Hilbert class field of E is $E(\sqrt{2})$.

The local condition $n \in N_{E/\mathbb{Q}}(\mathfrak{o}_{E_p})$ for all primes $p \leqslant \infty$ is equivalent to the condition (1). Since $\mathrm{Gal}(E/\mathbb{Q}) = \mathrm{Gal}(E_v/\mathbb{Q}_2)$, one concludes that $n \in N_{E/\mathbb{Q}}(E^\times)$ by [14, Theorem 6.11].

If $(x_p)_{p \leqslant \infty}$ is the local solution with $x_p \in \mathfrak{o}_{E_p}$ and $N_{E/\mathbb{Q}}(x_p) = n$, for all $p \leqslant \infty$, one can verify that $\psi_{H/E}((x_p)_{p \leqslant \infty}) = 1$. The computation of $\psi_{\Theta/L}[(N_{E/L}(x_p))_{p \leqslant \infty}]$ is the same as that in Example 5.5. □

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